Unit

17 Concentration of solutions and volumetric analysis

Unit Key Concepts

- Concentration (in g dm^{-3})
- Preparing standard solutions
- Volumetric analysis

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

17.1 Concentration of a solution

We have learnt in Unit 15 that the concentration of a solution can be expressed in mol dm^{-3}. We can also express the concentration of a solution in g dm^{-3}.

✓ Concentration of a solution (g dm^{-3}) = $\dfrac{\text{mass of solute (g)}}{\text{volume of solution } (dm^3)}$

Example 17.1

Q 8.40 g of sodium carbonate were dissolved in water and the solution was made up to 500.0 cm^3. Calculate the concentration of the solution prepared in g dm^{-3}.

A Volume of solution = $\dfrac{500.0}{1\,000}$ dm^3

Concentration of sodium carbonate solution

$$= \frac{\text{mass of } Na_2CO_3}{\text{volume of solution}}$$

$$= \frac{8.40 \text{ g}}{\left(\dfrac{500.0}{1\,000}\right) dm^3}$$

$$= 16.8 \text{ g } dm^{-3}$$

$\therefore$ the concentration of the sodium carbonate solution was 16.8 g dm^{-3}.

Example 17.2

Q The concentration of a calcium nitrate solution is 19.7 g dm^{-3}.

What is the molar concentration of the calcium nitrate solution?

(Relative atomic masses: N = 14.0, O = 16.0, Ca = 40.1)

A Molar mass of $Ca(NO_3)_2$ = [40.1 + 2 x (14.0 + 3 x 16.0)] g mol^{-1}
= 164.1 g mol^{-1}

Mass of $Ca(NO_3)_2$ in 1.00 dm^3 of solution = 19.7 g

$$\text{Number of moles of } Ca(NO_3)_2 \text{ in 1.00 } dm^3 \text{ of solution} = \frac{\text{mass}}{\text{molar mass}}$$
$$= \frac{19.7 \text{ g}}{164.1 \text{ g mol}^{-1}}$$
$$= 0.120 \text{ mol}$$

$$\text{Molar concentration of calcium nitrate solution} = \frac{\text{number of moles of } Ca(NO_3)_2}{\text{volume of solution}}$$
$$= \frac{0.120 \text{ mol}}{1.00 \text{ dm}^3}$$
$$= 0.120 \text{ mol dm}^{-3}$$

∴ the molar concentration of the calcium nitrate solution is 0.120 mol dm^{-3}.

CE 2009 Paper 1 Q4(c)
DSE 2014 Paper 1B Q7(a)

Example 17.3

Q A sample of concentrated nitric acid has a density of 1.37 g cm^{-3} and contains 65.4% of nitric acid by mass.

What is the concentration of the nitric acid

a) in g dm^{-3}?

b) in mol dm^{-3}?

(Relative atomic masses: H = 1.0, N = 14.0, O = 16.0)

Continued on next page

A1 a) Consider 1 000 cm^3 (i.e. 1.00 dm^3) of the sample.

Mass of 1 000 cm^3 of the sample = 1.37 g cm^{-3} x 1 000 cm^3
= 1 370 g

Mass of HNO_3 in 1 000 cm^3 of sample
= mass of 1 000 cm^3 of sample x percentage by mass of HNO_3 in sample
= 1 370 g x 65.4%
= 896 g

$$\text{Concentration of nitric acid in g dm}^{-3} = \frac{\text{mass of HNO}_3}{1.00\ \text{dm}^3}$$
$$= \frac{896\ \text{g}}{1.00\ \text{dm}^3}$$
$$= 896\ \text{g dm}^{-3}$$

b) Molar mass of HNO_3 = (1.0 + 14.0 + 3 x 16.0) g mol^{-1}
= 63.0 g mol^{-1}

$$\text{Number of moles of HNO}_3 \text{ in 1.00 dm}^3 \text{ of sample} = \frac{\text{mass}}{\text{molar mass}}$$
$$= \frac{896\ \text{g}}{63.0\ \text{g mol}^{-1}}$$
$$= 14.2\ \text{mol}$$

$$\text{Molarity of nitric acid} = \frac{\text{number of moles of HNO}_3}{\text{volume of solution}}$$
$$= \frac{14.2\ \text{mol}}{1.00\ \text{dm}^3}$$
$$= 14.2\ \text{mol dm}^{-3}$$

∴ the concentration of the nitric acid sample is 896 g dm^{-3} and 14.2 mol dm^{-3}.

Practice 17.1 Refer to page T36.

1 5.80 g of magnesium sulphate crystals ($MgSO_4 \bullet 7H_2O$) were used to prepare 250.0 cm^3 of solution. Calculate the concentration, in g dm^{-3}, of the solution obtained.
23.2 g dm^{-3}

2 A sample of concentrated phosphoric acid has a density of 1.25 g cm^{-3} and contains 81.8% of phosphoric acid by mass. What is the concentration of the phosphoric acid

a) in g dm^{-3}?
1 020 g dm^{-3}

b) in mol dm^{-3}?
10.4 mol dm^{-3}

(Relative atomic masses: H = 1.0, O = 16.0, P = 31.0)

17.2 Dilution

Concentrated acids are diluted by adding them slowly to a large amount of water while stirring.

To save space in the laboratory, commonly used solutions are often purchased in concentrated form. Water is added to achieve the molarity desired for a particular solution. This process is called **dilution**.

Since only water is added during the dilution, all the solute in the final dilute solution must come from the original concentrated solution, that is:

$$\begin{matrix}\text{number of moles of solute} \\ \text{before dilution}\end{matrix} = \begin{matrix}\text{number of moles of solute} \\ \text{after dilution}\end{matrix}$$

As the number of moles of solute in a solution = MV, where M = molarity and V = volume, therefore we have

✓ (MV) before dilution = (MV) after dilution

Example 17.4

Q A student added 200.0 cm^3 of distilled water to 600.0 cm^3 of 2.20 mol dm^{-3} nitric acid. What was the molarity of the diluted acid?

(Assume the volume of the solution obtained is 800.0 cm^3.)

CE 2009 Paper 2 Q35
CE 2011 Paper 1 Q9(b)

A (MV) before dilution = (MV) after dilution, where M = molarity, V = volume

$$2.20 \times \frac{600.0}{1\,000} = M \times \frac{800.0}{1\,000}$$

$$M = 1.65$$

$\therefore$ molarity of the diluted acid was 1.65 mol dm^{-3}.

dilution 稀釋

Example 17.5

Q What volume of water must be used to make 20.0 cm^3 of 7.50 mol dm^{-3} sodium hydroxide solution to exactly 2.00 mol dm^{-3}?

A Suppose V cm^3 of 2.00 mol dm^{-3} sodium hydroxide solution can be obtained.

(MV) before dilution = (MV) after dilution, where M = molarity, V = volume

$$7.50 \times \frac{20.0}{1\,000} = 2.00 \times \frac{V}{1\,000}$$

$$V = 75.0$$

Volume of dilute solution obtained = 75.0 cm^3

$\therefore$ volume of water used = (75.0 – 20.0) cm^3

= 55.0 cm^3

Practice 17.2 Refer to page T37.

1 40.0 cm^3 of 6.60 mol dm^{-3} sulphuric acid were mixed with 80.0 cm^3 of water. What was the molarity of the diluted acid?
2.20 mol dm^{-3}

2 What volume of water must be added to 50.0 cm^3 of 3.60 mol dm^{-3} sodium carbonate solution to make it exactly 1.50 mol dm^{-3}?
70.0 cm^3

17.3 Volumetric analysis

Volumetric analysis is commonly used in quantitative analysis.

Volumetric analysis is a method of finding out the concentration (or amount) of a substance in a sample. It relies on the precise measurement of volumes of solutions involved in chemical reactions.

In volumetric analysis, just sufficient volume of a solution of known concentration is allowed to completely react with the substance being analyzed in a sample. From the measured volume and known concentration of the solution used, we can calculate the concentration (or amount) of the substance in question.

The process of determining the 'just sufficient' volume is called titration. The process of determining the concentration of an acid or an alkali using an acid-alkali reaction is an acid-alkali titration.

volumetric analysis 容量分析

Apparatus used in volumetric analysis

Fig. 17.1 shows some special apparatus used in volumetric analysis.

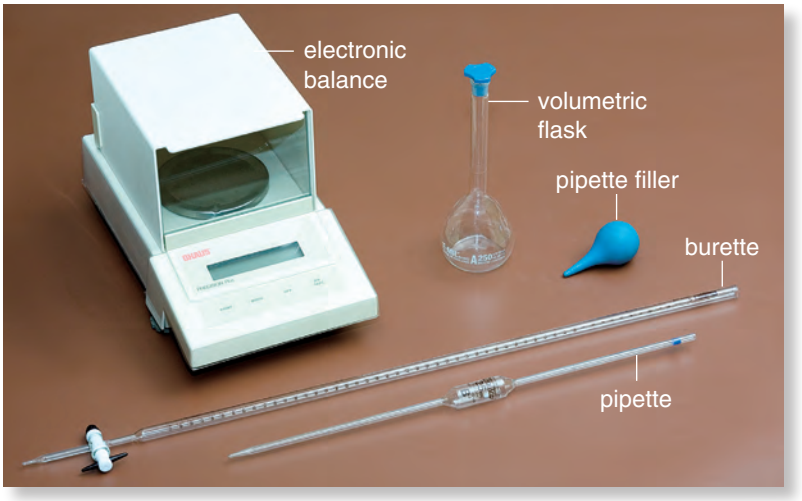


Fig. 17.1 Special apparatus used in volumetric analysis

- An **electronic balance** is a piece of fast and accurate equipment for measuring masses of chemicals (accuracy up to 0.001 g).
- A **volumetric flask** is a flask used to prepare a solution of accurately known volume (e.g. 250.0 cm^3).
- A **pipette** and a **pipette filler** are used to deliver a fixed accurate volume (e.g. 25.0 cm^3) of a solution.
- A **burette** is used to deliver various volumes of solution accurately.

DSE 2012 Paper 1B Q7(d)(i)

electronic balance 電子天平　volumetric flask 容量瓶　pipette 移液管　pipette filler 移液管膠泵　burette 滴定管

17.4 Preparing a standard solution of an acid / alkali

✓ A **standard solution** is a solution with an accurately known concentration. N1

N1 Students should be able to explain the meaning of a 'standard solution'.

CE 2007 Paper 1 Q10(d)(i)

Dissolving a solid acid / alkali in water

Video
Preparing 250.0 cm^3 of about 0.1 mol dm^{-3} ethanedioic acid solution by dissolving the solid acid

We can prepare a standard solution by dissolving a known mass of a solid acid / alkali in a known volume of solution. The solid should have the following characteristics:

- it must be obtainable in a very pure form;
- it must have a known chemical formula (including the number of molecules of water of crystallization in a *hydrated compound*);
- it should dissolve in water completely at room temperature;
- it must be stable and does not absorb moisture from the air;
- it should have a high molar mass to minimize weighing errors.

Few compounds are suitable for preparing standard solutions. Some common compounds, such as potassium hydroxide and concentrated sulphuric acid, absorb moisture from the air and cannot be weighed accurately.

Two solids suitable for preparing standard solutions are ethanedioic acid crystals ($(COOH)_2 \cdot 2H_2O$) and anhydrous sodium carbonate (Na_2CO_3). N2

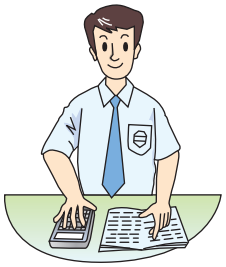
N2 Solid sodium hydroxide is NOT suitable for preparing a standard solution.

CE 2007 Paper 1 Q10(d)(ii)
DSE 2013 Paper 1B Q4(c)

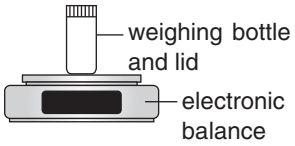
Fig. 17.2 shows the procedure for preparing 250.0 cm^3 of an approximately 1 mol dm^{-3} solution of a solid acid. N3

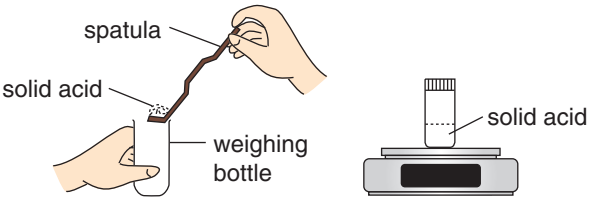
N3 When preparing a solution from a solid acid, a pipette is NOT required for transferring the acid solution to the volumetric flask.

standard solution 標準溶液　　hydrated compound 水合化合物

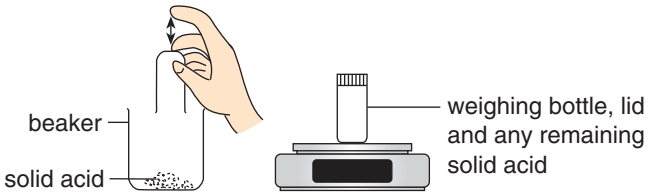


1 Calculate the mass of the solid acid required.

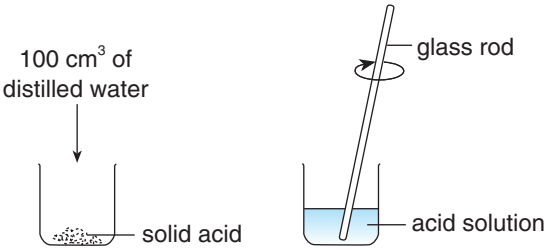




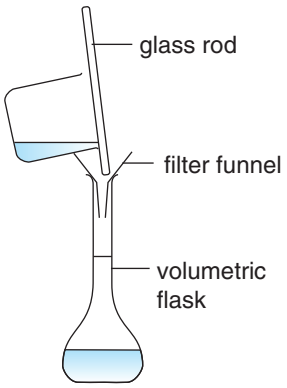
2 (a) Weigh an empty dry **weighing bottle** and lid accurately.
(b) Add the solid acid to the bottle until we have approximately the mass calculated.
(c) Weigh the weighing bottle, the lid and the solid acid accurately.



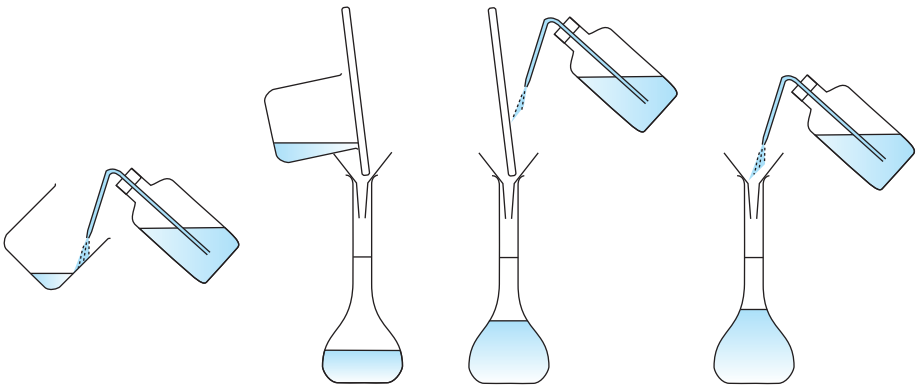
3 (a) Tip out the solid acid into a beaker as much as possible.
(b) Reweigh the weighing bottle, the lid and any remaining solid acid accurately.



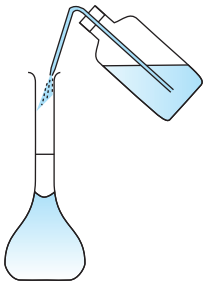
4 Add 100 cm^3 of distilled water to the beaker. Stir with a glass rod until all the solid acid has dissolved.



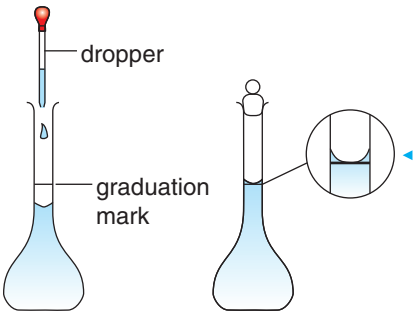
5 Transfer the solution into a 250.0 cm^3 volumetric flask with the aid of a filter funnel.



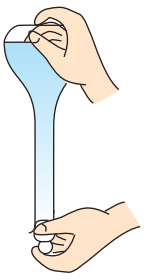
6 Wash the beaker, the glass rod and the filter funnel with a little distilled water several times. Pour all the washings into the flask.



7 Add distilled water to the flask until the **meniscus** is about 2 cm below the graduation mark.



8 Add distilled water using a dropper until the meniscus reaches the graduation mark.



9 Stopper the flask. Turn it upside down several times to mix the solution well.

Fig. 17.2 Steps for preparing 250.0 cm^3 of an approximately 1 mol dm^{-3} solution of a solid acid

DSE 2014 Paper 1B Q7(b)(i)

The eyes must be on the same level as the graduation mark.

weighing bottle 稱量瓶　　meniscus 彎液面

Suppose we use ethanedioic acid crystals ($(COOH)_2 \bullet 2H_2O$) to prepare 250.0 cm^3 of an approximately 1 mol dm^{-3} solution. Upon calculation, we know that approximately 31 g of the acid crystals are required.

The following results are obtained:

Mass of weighing bottle + lid = 23.9 g

Mass of weighing bottle + lid + acid crystals = 57.0 g

Mass of weighing bottle + lid + remaining acid crystals = 24.2 g

Mass of acid crystals used = (57.0 – 24.2) g
= 32.8 g

Molar mass of $(COOH)_2 \bullet 2H_2O$
= [2 x (12.0 + 2 x 16.0 + 1.0) + 2 x (2 x 1.0 + 16.0)] g mol^{-1}
= 126.0 g mol^{-1}

Number of moles of $(COOH)_2 \bullet 2H_2O$ used

$$= \frac{\text{mass}}{\text{molar mass}}$$

$$= \frac{32.8 \text{ g}}{126.0 \text{ g mol}^{-1}}$$

= 0.260 mol

Concentration of the acid solution

$$= \frac{\text{number of moles of } (COOH)_2 \bullet 2H_2O}{\text{volume of solution}}$$

$$= \frac{0.260 \text{ mol}}{\left(\frac{250.0}{1\,000}\right) \text{dm}^3}$$

= 1.04 mol dm^{-3}

Diluting a concentrated acid / alkali of known concentration

Some solid acids / alkalis cannot be weighed accurately. Hence we cannot prepare standard solutions by dissolving them in water. In this case, we dilute a concentrated acid / alkali of known concentration to the concentration required.

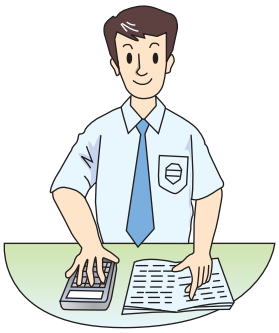
Video

Preparing 250.0 cm^3 of 0.10 mol dm^{-3} (M) sulphuric acid by diluting 1.0 mol dm^{-3} (M) sulphuric acid

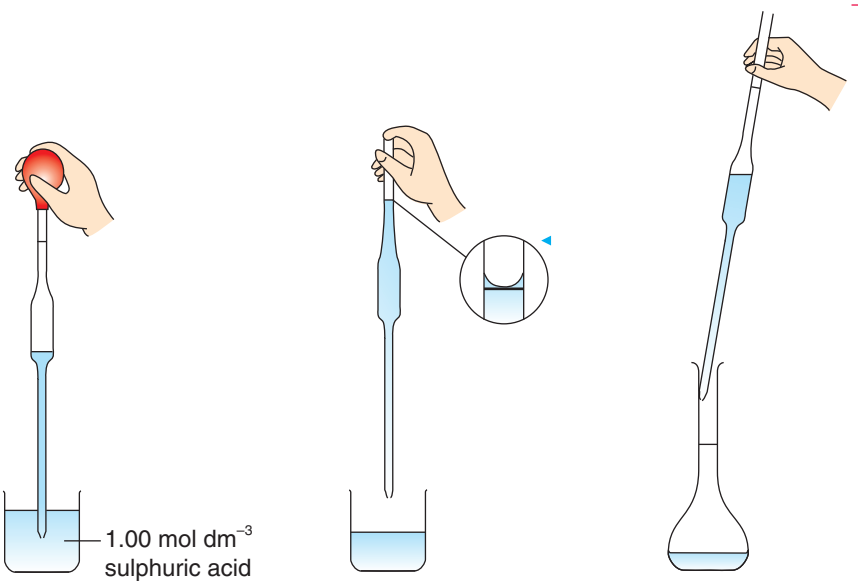
N4 Examination questions often ask about the procedure of diluting a concentrated acid / alkali of known concentration.

Fig. 17.3 shows the procedure for preparing 250.0 cm^3 of 0.100 mol dm^{-3} sulphuric acid by diluting 1.00 mol dm^{-3} sulphuric acid. N4

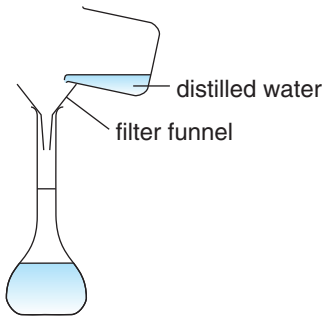
The eyes must be on the same level as the graduation mark.



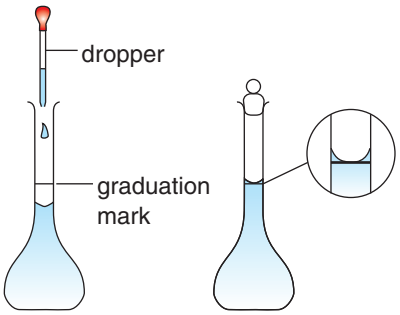
1 Calculate the volume of the original acid needed for dilution. 25.0 cm^3 of the original acid are required.



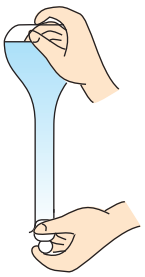
2 Deliver exactly 25.0 cm^3 of the original acid into a 250.0 cm^3 volumetric flask using a pipette and pipette filler.



3 Add distilled water to the flask until the meniscus is about 2 cm below the graduation mark.



4 Add distilled water using a dropper until the meniscus reaches the graduation mark.



5 Stopper the flask. Turn it upside down several times to mix the solution well.

Fig. 17.3 Steps for preparing an acid solution by dilution

CE 2007 Paper 1 Q10(a)
CE 2009 Paper 1 Q11(b)
CE 2010 Paper 2 Q19
CE 2011 Paper 1 Q9(a)

17.1
Preparing solutions of known concentrations.

Video
Preparing solutions of known concentrations

Assessment Form
An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

N5 Refer to page T37.

17.5 Acid-alkali titration

Let us consider the titration of an acidic solution of unknown concentration (solution A) with an alkaline solution of known concentration (solution B). Fig. 17.4 shows the experimental set-up of such a titration in the laboratory and Fig. 17.5 shows the procedure for an acid-alkali titration. N5

Reference Website
Refer to the following website for simulations of acid-alkali titrations:
http://www.vias.org/simulations/simusoft_titration.html
(accessed July 2014)

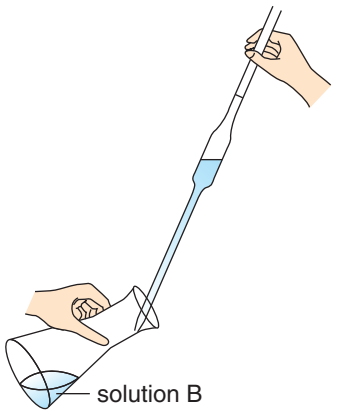
Video
Titration of dilute hydrochloric acid against dilute sodium hydroxide solution



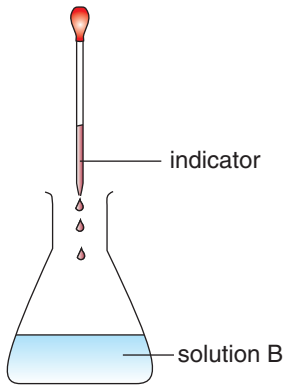
(a) Start of titration (b) End of titration

Fig. 17.4 Experimental set-up of an acid-alkali titration in the laboratory

The indicator is added so that a colour change will occur when the reaction between the acidic solution and the alkaline solution is complete. We will further discuss indicators later in this unit.



1 Deliver 25.0 cm^3 of solution B to a conical flask using a pipette.

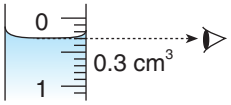


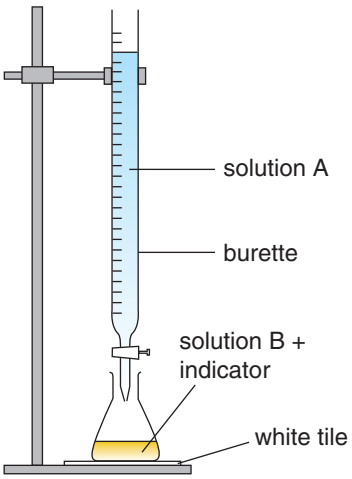
2 Add a few drops of an acid-alkali indicator to solution B.

CE 2006 Paper 1 Q9(a)

N6 Students should be able to draw a diagram for titration set-up.

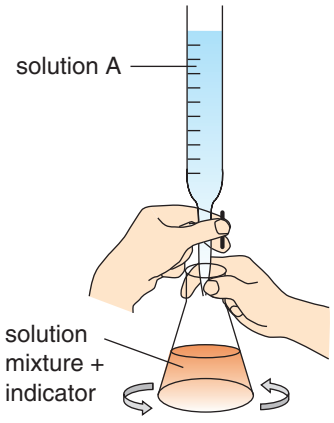
The eyes must be on the same level as the meniscus when taking the reading.





3 Fill a burette with solution A. Arrange the apparatus as shown. Record the initial burette reading. N6

CE 2011 Paper 1 Q9(c)



4 Run solution A from the burette into solution B until the reaction mixture just changes colour and shows that solution B is neutralized. Record the final burette reading.

Fig. 17.5 Steps in an acid-alkali titration

The amount of solution delivered by the burette is called a **titre**. To improve accuracy, repeat the titration until three titres differing within 0.1 cm^3 are obtained. These are known as consistent results and are used to determine the mean titre. Record the results in a table as shown below.

Volume of solution A required to react completely with 25.0 cm^3 of solution B				
Burette reading (cm^3)	**Titration**			
	1 (trial)	**2**	**3**	**4**
Final burette reading				
Initial burette reading				
Volume required				

Fig. 17.6 shows a flow diagram which summarizes the procedure for an acid-alkali titration.

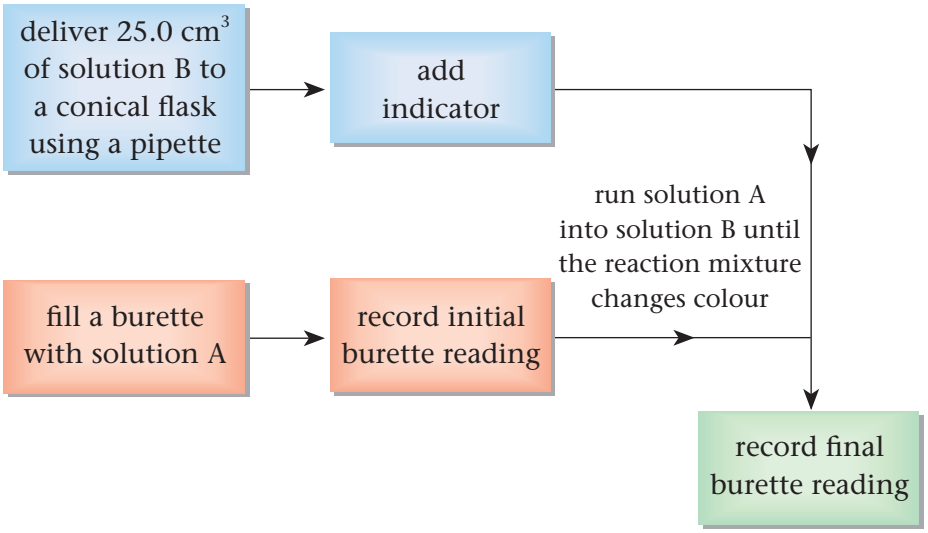


Fig. 17.6 A flow diagram showing the procedure for an acid-alkali titration

Using the experimental data obtained and the chemical equation for the reaction, we can find the concentration of the unknown acidic solution, as illustrated in the following example.

titre 滴定值

Example 17.6

Q In a certain experiment, four 25.0 cm^3 portions of 0.456 mol dm^{-3} sodium hydroxide solution were titrated with sulphuric acid. The results of the experiment are shown in the following table.

Burette reading (cm^3)	Titration			
	1st (trial)	2nd	3rd	4th
Final reading	31.40	28.55	30.50	29.55
Initial reading	0.00	0.00	2.00	1.05

a) Calculate a reasonable average for the volume of sulphuric acid required to neutralize 25.0 cm^3 of the alkali completely.

b) Determine the molarity of the sulphuric acid.

A a) The volumes of sulphuric acid required for neutralization were 31.40 cm^3, 28.55 cm^3, 28.50 cm^3 and 28.50 cm^3. It is reasonable to discard the first reading since it is very different from the other three and it is the result obtained in the first trial.

$\therefore$ average volume of sulphuric acid required for neutralization

$$= \frac{28.55 + 28.50 + 28.50}{3} \text{ cm}^3$$

$$= 28.52 \text{ cm}^3$$

b) $2NaOH(aq) + H_2SO_4(aq) \longrightarrow Na_2SO_4(aq) + 2H_2O(l)$

0.456 mol dm^{-3} ? mol dm^{-3}

25.0 cm^3 28.52 cm^3

Number of moles of NaOH in 25.0 cm^3 solution = molarity of solution x volume of solution

$$= 0.456 \text{ mol dm}^{-3} \times \frac{25.0}{1\,000} \text{ dm}^3$$

$$= 0.0114 \text{ mol}$$

According to the equation, 2 moles of NaOH require 1 mole of H_2SO_4 for complete neutralization.

i.e. number of moles of H_2SO_4 in 28.52 cm^3 solution $= \frac{0.0114}{2}$ mol

$$= 0.00570 \text{ mol}$$

$$\text{Molarity of sulphuric acid} = \frac{\text{number of moles of } H_2SO_4}{\text{volume of solution}}$$

$$= \frac{0.00570 \text{ mol}}{\left(\frac{28.52}{1\,000}\right) \text{ dm}^3}$$

$$= 0.200 \text{ mol dm}^{-3}$$

$\therefore$ the molarity of the sulphuric acid is 0.200 mol dm^{-3}.

Fig. 17.7 summarizes the steps we use in the above calculation.

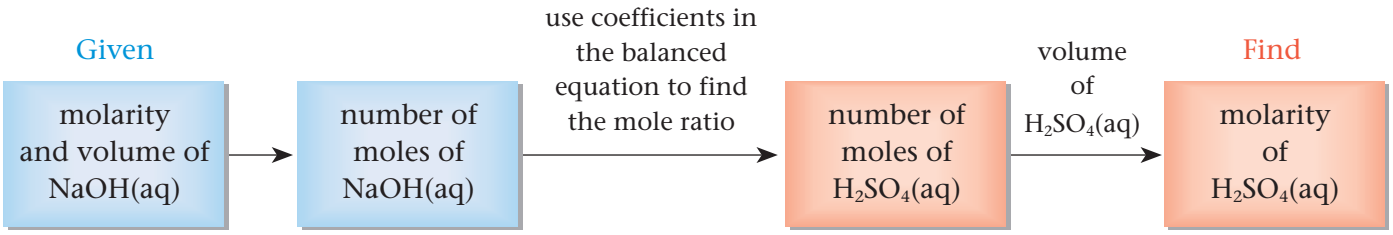


Fig. 17.7 Steps in calculating the concentration of an acid from titration data

Washing apparatus in volumetric analysis

For results of a volumetric analysis to be reliable and accurate, it is essential to wash the apparatus correctly (Table 17.1).

If we wash the pipette or burette with distilled water only, water droplets remaining on the inside of the glassware will dilute the solution that the glassware is going to contain.

Table 17.1

Washing apparatus in volumetric analysis

Wash with \ Apparatus	Pipette	Conical flask	Burette	Volumetric flask
distilled water	✔	✔	✔	✔
the solution it is going to contain	✔		✔	

DSE 2013 Paper 1B Q4(e)(i)

CE 2006 Paper 2 Q47

Worksheet

A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

A student proposed a method to determine the concentration of ethanoic acid in a sample of vinegar by titration with standard sodium hydroxide solution. The method proposed consists of the following steps:

Step 1 Prepare a standard sodium hydroxide solution by dissolving a known mass of sodium hydroxide pellets in distilled water and then making it up to 250.0 cm^3.

Step 2 Rinse a conical flask with a little of the sample of vinegar. Then transfer 25.0 cm^3 of the sample to the flask.

Step 3 Fill a burette with the sodium hydroxide solution.

Step 4 Titrate the sample in the conical flask with the sodium hydroxide solution.

Step 5 Using this titration result, calculate the concentration of ethanoic acid in the sample.

Point out THREE inappropriate practices in the method. Explain why they are inappropriate.

Discussion

Step 1 A standard sodium hydroxide solution should not be prepared using the method described.
Explanation: Sodium hydroxide pellets absorb moisture and carbon dioxide from the air.

Step 2 The conical flask should not be rinsed with the sample of vinegar.
Explanation: This would increase the amount of sample in the flask.

Step 5 Calculation should not be based on the result of one titration only.
Explanation: There may be errors in the titration.

17.6 pH change during a titration

17.2
Following the pH change during an acid-alkali titration.

The point in an acid-alkali titration at which the reactants just react completely with each other is called the **equivalence point**.

Consider the reaction between dilute hydrochloric acid and dilute sodium hydroxide solution: N7

$$NaOH(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l)$$

At the equivalence point, the acid and the alkali just react completely with each other. The pH of the solution mixture is 7.00.

N7 Consider an aqueous solution of ethanoic acid with a pH value of 4.
Adding
- solid calcium carbonate; and
- dilute aqueous ammonia

would increase the pH value of the acid.

Video
Following the pH change during an acid-alkali titration

Suppose we deliver the dilute sodium hydroxide solution into a reaction vessel and insert the electrode of a pH meter into the solution as shown in Fig. 17.8. Fig. 17.9 shows the pH change of the solution mixture as we add a standard hydrochloric acid to it. This is a **titration curve**.

- When the acid is first added, the pH decreases very slightly — the alkali is in great excess.
- Within 1–2 cm^3 of the equivalence point, the pH starts to decrease more quickly. Eventually, there is a very sharp decrease in pH brought about by a very small addition of the acid. This shows on the titration curve as a vertical part.
- As further acid is added, there is little additional decrease in pH — the acid is now in great excess.

Animation
A strong acid-strong alkali titration curve

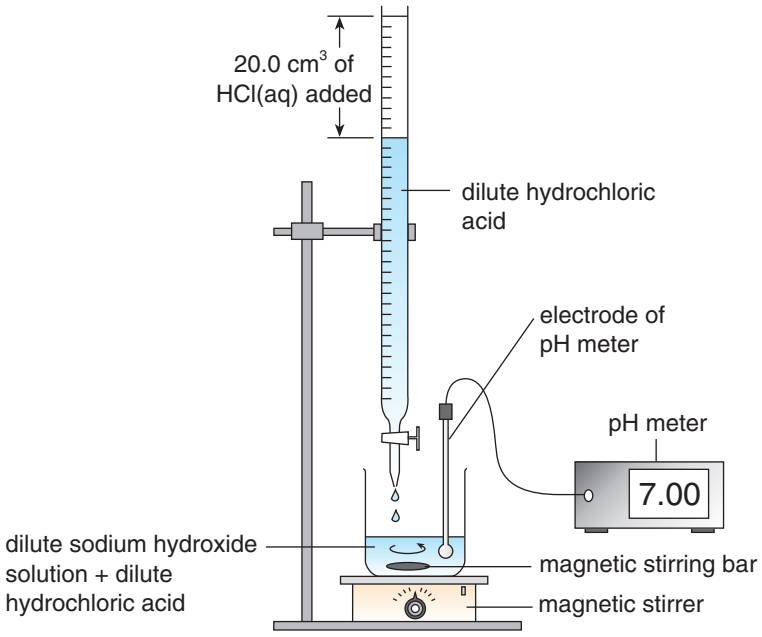


Fig. 17.8 Experimental set-up for monitoring the pH change when dilute hydrochloric acid is added to dilute sodium hydroxide solution

equivalence point 當量點 titration curve 滴定曲線

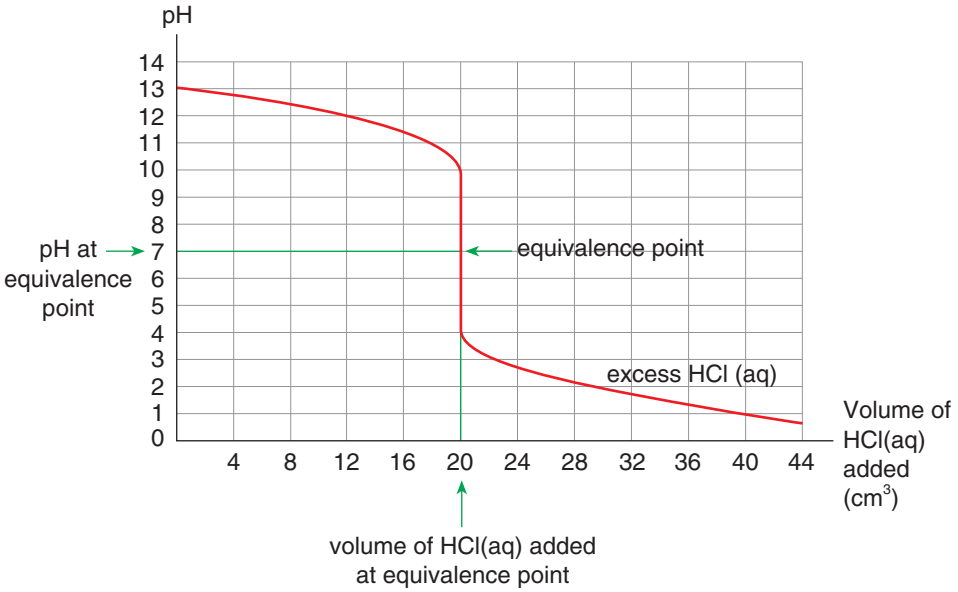


Fig. 17.9 The titration curve obtained when dilute hydrochloric acid is added to dilute sodium hydroxide solution

From the titration curve, we can obtain the volume of acid required to react completely with the sodium hydroxide solution in the reaction vessel. Then we can calculate the concentration of the sodium hydroxide solution.

17.7 Using an indicator in an acid-alkali titration

If we do not use a pH meter to follow an acid-alkali titration, we may use an acid-alkali indicator to detect the equivalence point.

The point at which the indicator changes colour is known as the **end point**. The colour change of an indicator must be sharp rather than gradual at the end point, i.e. no more than one drop of acid (or alkali) is needed to give a complete colour change.

Figs. 17.10–17.11 show the colours of two common acid-alkali indicators at different pH values.

The end point and the equivalence point usually occur at slightly different times during a titration. However, we have to assume that the end point is the equivalence point and recognize this assumption as a source of error.

end point 終點

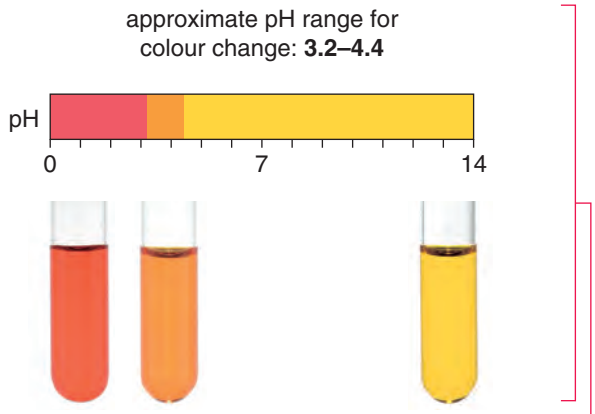


Fig. 17.10 Colours of methyl orange at different pH values

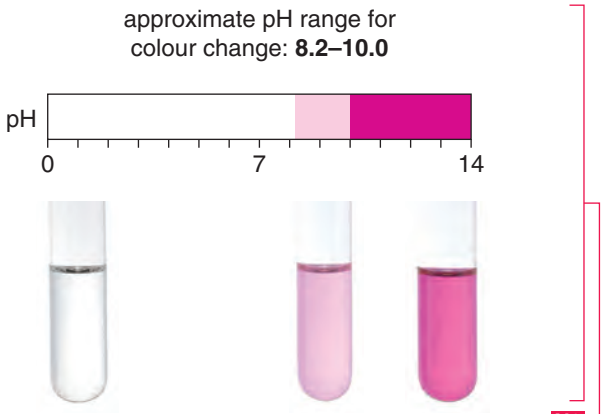


Fig. 17.11 Colours of phenolphthalein at different pH values

N8

CE 2011 Paper 1 Q9(d)
ASL 2009 Paper 2 Q3(a)
DSE 2012 Paper 1B Q7(d)(ii)

CE 2005 Paper 2 Q40
DSE 2013 Paper 1B Q4(d)(i)

N8 Colour change of phenolphthalein in an acid-alkali titration

Consider the titration of 25.0 cm^3 of a dilute acid with sodium hydroxide solution using phenolphthalein as an indicator.

At the beginning of the titration, the solution in the conical flask turns pink upon the addition of sodium hydroxide solution. But it becomes colourless immediately upon swirling. However, near the end point, the solution takes a longer time to become colourless upon swirling.

The time needed for the solution to become colourless is different at the two stages.

Neutralization is a quick process. As the titration proceeds, concentration of the acid decreases, hence the rate of reaction decreases.

Animation
Titration curves

Choosing a suitable indicator for an acid-alkali titration

In an acid-alkali titration, the pH of the solution mixture in the conical flask varies with the volume of solution added from the burette. The way in which the pH changes during a titration depends on the strengths of the acid and alkali used. In general there are four types of acid-alkali titration:

- strong acid-strong alkali titration;
- weak acid-strong alkali titration;
- strong acid-weak alkali titration;
- weak acid-weak alkali titration.

Fig. 17.12 shows the pH ranges within which methyl orange and phenolphthalein change colours. A suitable indicator for a titration must change colour within the pH range of the vertical part of the titration curve.

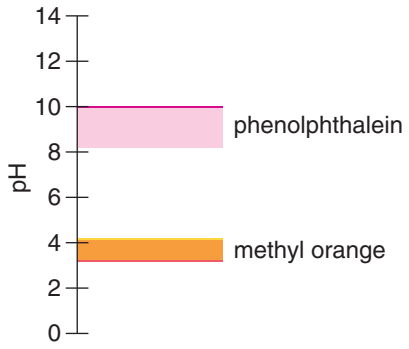


Fig. 17.12 pH colour ranges for two common indicators

Figs. 17.13 and 17.14 show a strong acid-strong alkali titration curve and a weak acid-strong alkali titration curve respectively. There is often a marked change in pH at the equivalence point.

Consider the strong acid-strong alkali titration curve shown in Fig. 17.13. Both methyl orange and phenolphthalein are suitable indicators for the titration because both indicators change colour within the pH range of the vertical part of the titration curve.

CE 2009 Paper 1 Q11(c)

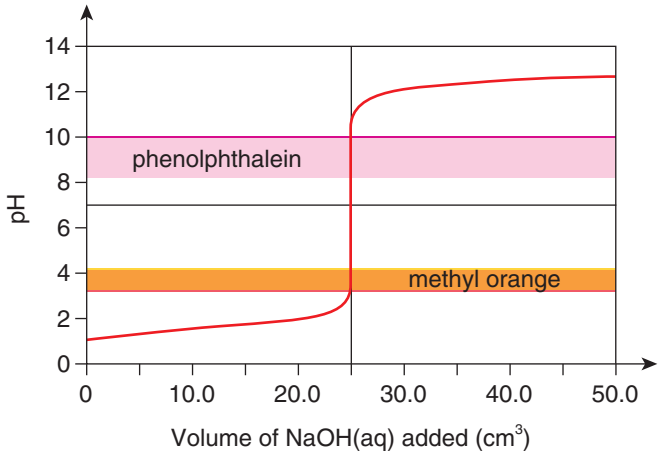


Fig. 17.13 A strong acid-strong alkali titration curve (for the titration of 25.0 cm^3 of 0.100 mol dm^{-3} HCl(aq) with 0.100 mol dm^{-3} NaOH(aq))

Consider the weak acid-strong alkali titration curve shown in Fig. 17.14. Phenolphthalein is a suitable indicator for the titration. On the other hand, methyl orange does not change colour within the pH range of the vertical part of the curve. Thus, it is not a suitable indicator for such a titration.

ASL 2006 Q12(b)

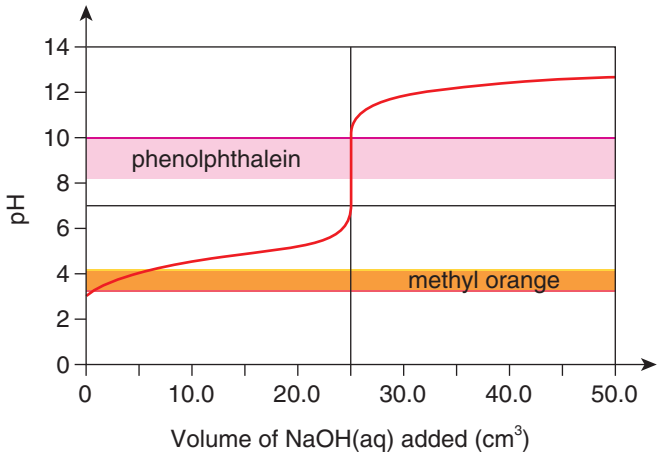


Fig. 17.14 A weak acid-strong alkali titration curve (for the titration of 25.0 cm^3 of 0.100 mol dm^{-3} CH_3COOH(aq) with 0.100 mol dm^{-3} NaOH(aq)) N9

N9 Examination questions may ask students to sketch the pH change for the titration of a weak acid (e.g. propanoic acid) with dilute NaOH(aq).

Fig. 17.15 shows a strong acid-weak alkali titration curve. Methyl orange is a suitable indicator for the titration. Phenolphthalein does not change colour within the pH range of the vertical part of the curve. Thus, it is not a suitable indicator for such a titration.

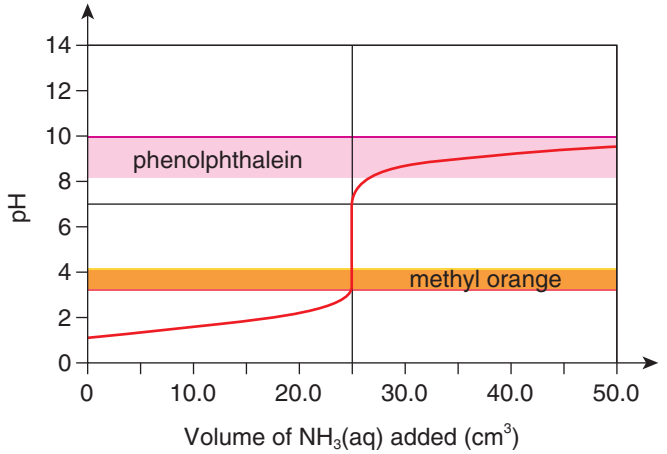


Fig. 17.15 A strong acid-weak alkali titration curve (for the titration of 25.0 cm^3 of 0.100 mol dm^{-3} HCl(aq) with 0.100 mol dm^{-3} NH_3(aq))

Fig. 17.16 shows a weak acid-weak alkali titration curve. We can see that there is no significant pH change at the equivalence point. Thus, no indicator is suitable for such a titration.

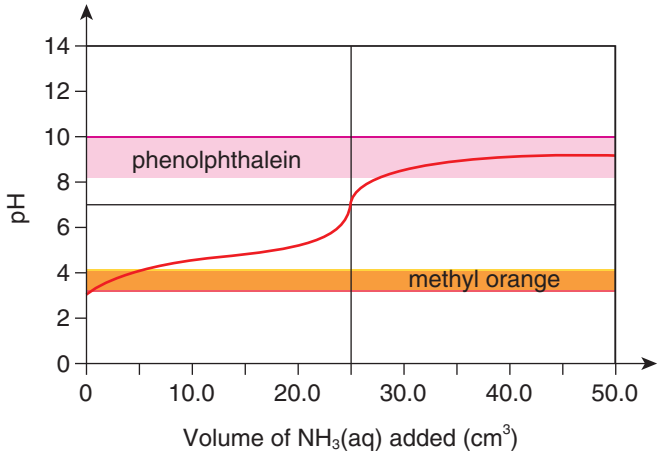


Fig. 17.16 A weak acid-weak alkali titration curve (for the titration of 25.0 cm^3 of 0.100 mol dm^{-3} CH_3COOH(aq) with 0.100 mol dm^{-3} NH_3(aq))

Table 17.2 lists the suitable indicators for different types of acid-alkali titration.

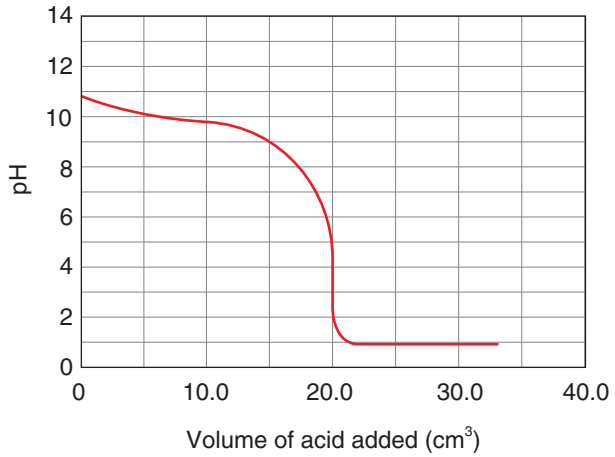
Table 17.2

Suitable indicators for different types of acid-alkali titrations

Type of acid-alkali titration	Example	Suitable indicator
Strong acid-strong alkali	HCl(aq) and NaOH(aq)	methyl orange or phenolphthalein
Weak acid-strong alkali	CH_3COOH(aq) and NaOH(aq)	phenolphthalein
Strong acid-weak alkali	HCl(aq) and NH_3(aq)	methyl orange
Weak acid-weak alkali	CH_3COOH(aq) and NH_3(aq)	—

Practice 17.3

The graph below shows the variation of pH of the solution mixture when 25.0 cm^3 of 0.100 mol dm^{-3} NH_3(aq) are titrated against an acid.



The table below lists the pH ranges for colour change of three acid-alkali indicators:

Indicator	pH range
Methyl orange	3.2 to 4.4
Bromothymol blue	6.0 to 7.6
Phenolphthalein	8.2 to 10.0

Choose from the above table a suitable indicator for the titration. Explain your answer.

Methyl orange is a suitable indicator as the indicator changes colour within the pH range of the vertical part of the titration curve.

N10 Refer to page T37.

17.8 Equivalence point detection by temperature change N10

Heat is released in a neutralization reaction. Besides using acid-alkali indicators, we can detect the equivalence point of a titration by the temperature change of the solution mixture. The temperature of the solution mixture is the highest at the equivalence point.

For example, consider the addition of 2 mol dm^{-3} hydrochloric acid into a polystyrene cup containing 25.0 cm^3 of 2 mol dm^{-3} sodium hydroxide solution (Fig. 17.17). The acid is added in 5 cm^3 portions. Fig. 17.18 shows the temperature change during the titration.

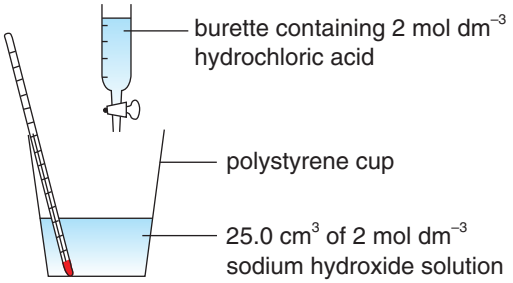


Fig. 17.17 Set-up for equivalence point detection by temperature change

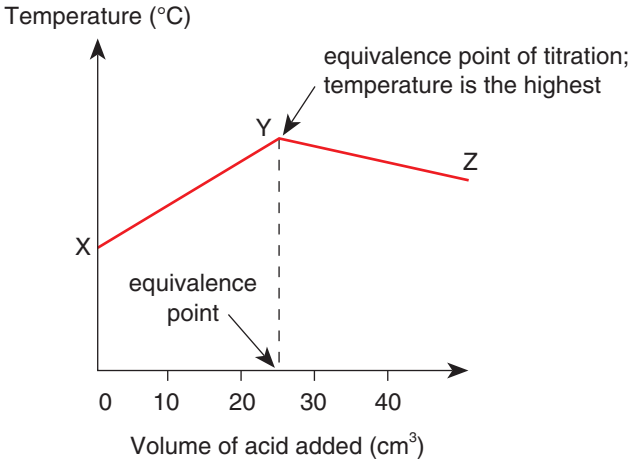


Fig. 17.18 Temperature change during the titration of sodium hydroxide solution with hydrochloric acid

Refer to Fig. 17.18. When dilute hydrochloric acid is added, it reacts with the sodium hydroxide solution. Heat is released. Thus, the temperature of the solution mixture rises from X to Y. Neutralization is completed at Y. No more heat is produced. The excess acid added also cools the solution mixture. Thus, the temperature of the solution mixture falls from Y to Z.

Assessment Form
Assessment forms on these activities are available for download from *Jing Kung Chemistry Website*.

17.3 **Assessment Form**
Determining the concentration of citric acid in lemon squash.

17.4 **Assessment Form**
Determining the concentration of ammonia in a household glass cleanser.

N11 Suppose a solution containing 3 moles of KOH reacts with another solution containing 1 mole of an acid for complete neutralization. It can be deduced that 1 mole of the acid provides 3 moles of $H^+(aq)$ ions.

N12 Refer to page T39.

17.9 Steps to solve problems of acid-alkali titrations

To solve problems of acid-alkali titrations, we can follow the steps below.

1 Write down a balanced chemical equation for the reaction.

$aA + bB \longrightarrow cC + dD$

2 Write down the amounts of substances given in the question underneath their formulae. **N11** **N12**

3 Suppose the amount of A is given. Convert this to number of moles.

$$\text{Number of moles of A} = \frac{\text{mass of A (g)}}{\text{molar mass of A (g mol}^{-1}\text{)}}$$

$$\text{or} = \text{molarity (mol dm}^{-3}\text{)} \times \text{volume of solution (dm}^3\text{)}$$

4 Using the balanced chemical equation, work out the number of moles of the substance (i.e. B, C or D) required in the question.

5 Convert the number of moles of the substance (i.e. B, C or D) required to the desired unit, for example, gram, molarity and cubic decimetre (dm^3).

Mass of substance (g)
= number of moles (mol) x molar mass (g mol^{-1})

$$\text{Molarity of solution (mol dm}^{-3}\text{)} = \frac{\text{number of moles of solute (mol)}}{\text{volume of solution (dm}^3\text{)}}$$

$$\text{Volume of solution (dm}^3\text{)} = \frac{\text{number of moles of solute (mol)}}{\text{molarity of solution (mol dm}^{-3}\text{)}}$$

Fig. 17.19 summarizes the steps mentioned above.

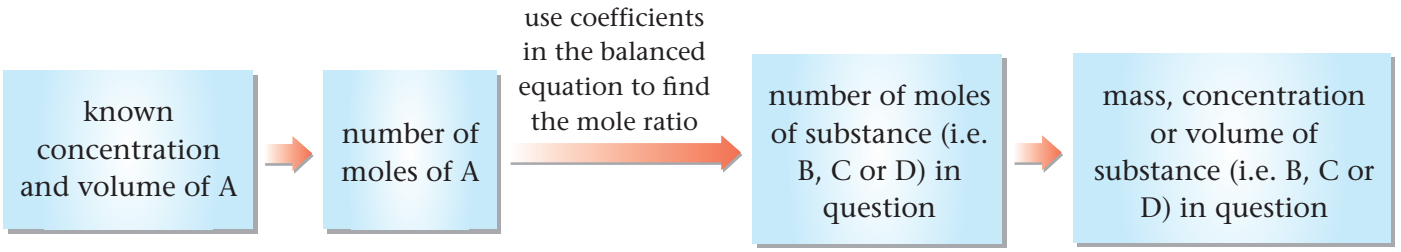


Fig. 17.19 Steps in solving problems of acid-alkali titrations

Example 17.7

Q An experiment was carried out to determine the concentration of a sodium hydroxide solution. 25.0 cm^3 of 1.80 mol dm^{-3} hydrochloric acid were diluted with distilled water to 250.0 cm^3. After that, 25.0 cm^3 of the dilute acid were titrated with the sodium hydroxide solution, using phenolphthalein as the indicator. 28.1 cm^3 of the sodium hydroxide solution were required to reach the end point.

a) State the colour change of the indicator at the end point of the titration.

b) Calculate the concentration of the diluted hydrochloric acid.

c) Calculate the concentration of the sodium hydroxide solution.

CE 2011 Paper 1 Q9(b)&(f)

DSE 2014 Paper 1B Q7(b)(ii)

A a) From colourless to pink

b) (MV) before dilution = (MV) after dilution, where M = molarity, V = volume

$$1.80 \times \frac{25.0}{1\,000} = M \times \frac{250.0}{1\,000}$$

$$M = 0.180$$

$\therefore$ the concentration of the diluted hydrochloric acid is 0.180 mol dm^{-3}.

c) 1 $HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$

2 0.180 mol dm^{-3} ? mol dm^{-3}

25.0 cm^3 28.1 cm^3

3 Number of moles of HCl in 25.0 cm^3 diluted acid = molarity of acid x volume of solution

$$= 0.180 \text{ mol dm}^{-3} \times \frac{25.0}{1\,000} \text{ dm}^3$$

$$= 0.00450 \text{ mol}$$

4 According to the equation, 1 mole of NaOH requires 1 mole of HCl for complete neutralization.

i.e. number of moles of NaOH in 28.1 cm^3 solution = 0.00450 mol

5 Concentration of sodium hydroxide solution $= \dfrac{\text{number of moles of NaOH}}{\text{volume of solution}}$

$$= \frac{0.00450 \text{ mol}}{\left(\frac{28.1}{1\,000}\right) \text{dm}^3}$$

$$= 0.160 \text{ mol dm}^{-3}$$

$\therefore$ the concentration of the sodium hydroxide solution is 0.160 mol dm^{-3}.

Practice 17.4

In an experiment, several 25.0 cm^3 portions of a calcium hydroxide solution were withdrawn using apparatus X. Each portion was titrated with 0.100 mol dm^{-3} hydrochloric acid, using a suitable indicator. The mean titre was 12.1 cm^3.

a) Name apparatus X.
Pipette

b) Write the chemical equation for the reaction between calcium hydroxide solution and hydrochloric acid.
$Ca(OH)_2(aq) + 2HCl(aq) \longrightarrow CaCl_2(aq) + 2H_2O(l)$

c) Suggest a suitable indicator for the titration.
Methyl orange

d) Calculate the concentration of the calcium hydroxide solution.
0.0242 mol dm^{-3}; refer to page T40.

Example 17.8

Q X is an acid. 25.0 cm^3 of 0.240 mol mol^{-3} solution X require 30.0 cm^3 of 0.600 mol dm^{-3} sodium hydroxide solution for complete neutralization. What is the basicity of X?

CE 2009 Paper 2 Q10

A Let n be the basicity of the acid, so we can represent the acid by H_nY.

$H_nY(s) + nNaOH(aq) \longrightarrow Na_nY(aq) + nH_2O(l)$

Number of moles of acid in 25.0 cm^3 of solution = molarity of solution x volume of solution

$$= 0.240 \text{ mol } dm^{-3} \times \frac{25.0}{1\,000} dm^3$$

$$= 0.00600 \text{ mol}$$

Number of moles of NaOH in 30.0 cm^3 solution = molarity of solution x volume of solution

$$= 0.600 \text{ mol } dm^{-3} \times \frac{30.0}{1\,000} dm^3$$

$$= 0.0180 \text{ mol}$$

$$\frac{\text{Number of moles of NaOH}}{\text{Number of moles of X}} = n$$

$$= \frac{0.0180 \text{ mol}}{0.00600 \text{ mol}}$$

$$n = 3$$

$\therefore$ the basicity of X is 3.

Example 17.9

Q A student was given a sample of a water-soluble metal carbonate, $M_2CO_3(s)$. In order to deduce what M was, the student prepared a 100.0 cm^3 aqueous solution of the carbonate using 1.66 g of the sample. The student then withdrew several 10.0 cm^3 portions of the solution and titrated each portion with 0.0750 mol dm^{-3} HCl(aq) using methyl orange as the indicator. The mean titre was 32.0 cm^3.

a) State the expected colour change of the indicator at the end point.

b) Based on the experimental results, determine the formula mass of M_2CO_3 and deduce what M is.

(Relative atomic masses: C = 12.0, O = 16.0)

A a) From yellow to orange

b) $M_2CO_3(s) + 2HCl(aq) \longrightarrow 2MCl(aq) + H_2O(l) + CO_2(g)$

1.66 g — 0.0750 mol dm^{-3}

↓ — 32.0 cm^3

100.0 cm^3

↓

(used) 10.0 cm^3

Number of moles of HCl in 32.0 cm^3 solution = molarity of solution x volume of solution

$$= 0.0750 \text{ mol dm}^{-3} \times \frac{32.0}{1\,000} \text{ dm}^3$$

$$= 0.00240 \text{ mol}$$

According to the equation, 1 mole of M_2CO_3 requires 2 moles of HCl for reaction.

i.e. number of moles of M_2CO_3 in 10.0 cm^3 solution $= \frac{0.00240}{2}$ mol

$= 0.00120$ mol

Number of moles of M_2CO_3 in 100.0 cm^3 solution $= 10 \times 0.00120$ mol

$= 0.0120$ mol

= number of moles of M_2CO_3 in 1.66 g

$$\therefore \frac{1.66 \text{ g}}{\text{formula mass of } M_2CO_3} = 0.0120 \text{ mol}$$

$$\text{formula mass of } M_2CO_3 = \frac{1.66 \text{ g}}{0.0120 \text{ mol}}$$

$$= 138 \text{ g mol}^{-1}$$

Let x be the relative atomic mass of M.

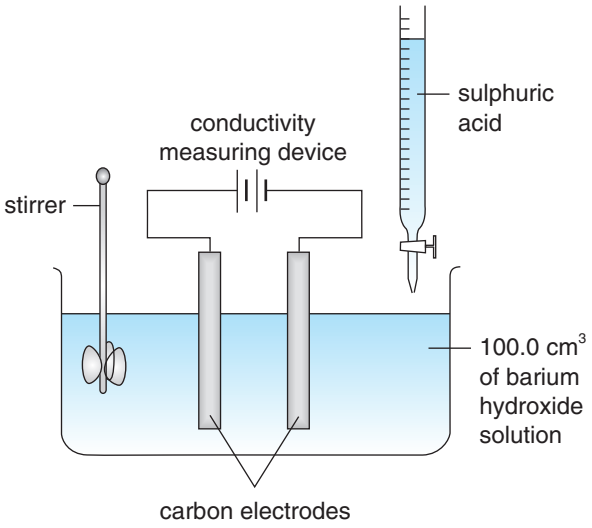
$2x + 12.0 + 3 \times 16.0 = 138$

$x = 39.0$

$\therefore$ M is likely to be potassium.

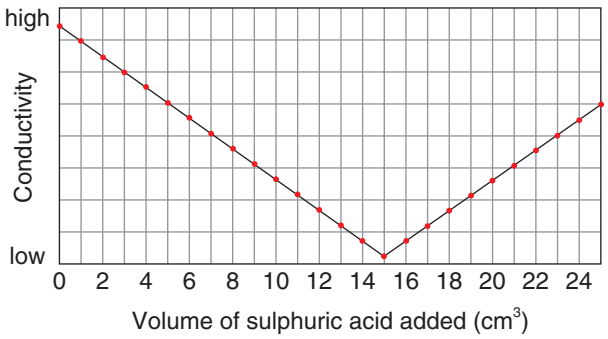
Example 17.10

Q Barium sulphate, a white precipitate, can be made by the reaction between barium hydroxide solution and sulphuric acid. The progress of the reaction between 100.0 cm^3 of barium hydroxide solution and sulphuric acid can be followed by measuring the electrical conductivity of the reaction mixture.



During the experiment, the sulphuric acid was added 1.0 cm^3 at a time. The conductivity was measured after each addition.

The results obtained are shown in the graph below.



a) Why did the electrical conductivity of the reaction mixture fall to almost zero? Explain your answer with the help of an appropriate equation.

b) What was the volume of sulphuric acid required to neutralize 100.0 cm^3 of barium hydroxide solution?

ASL 2010 Paper 1 Q9
AL 2010 Paper 1 Q7(a)

c) The concentration of the sulphuric acid was 1.00 mol dm^{-3}. What was the concentration of the barium hydroxide solution?

Continued on next page

A1 a) As sulphuric acid was added, it removed both the barium ions (by precipitation) and hydroxide ions (by neutralization).

$Ba(OH)_2(aq) + H_2SO_4(aq) \longrightarrow BaSO_4(s) + 2H_2O(l)$

At the equivalence point, all the barium ions and hydroxide ions had been removed. Hence the electrical conductivity of the reaction mixture fell to almost zero.

DSE 2014 Paper 1A Q7

b) $15.0\ cm^3$

c) $Ba(OH)_2(aq) + H_2SO_4(aq) \longrightarrow BaSO_4(s) + 2H_2O(l)$

? mol dm^{-3} 1.00 mol dm^{-3}

100.0 cm^3 15.0 cm^3

Number of moles of H_2SO_4 in 15.0 cm^3 solution = molarity of solution x volume of solution

$$= 1.00\ mol\ dm^{-3} \times \frac{15.0}{1\,000}\ dm^3$$

$$= 0.0150\ mol$$

According to the equation, 1 mole of $Ba(OH)_2$ requires 1 mole of H_2SO_4 for complete neutralization.

i.e. number of moles of $Ba(OH)_2$ in 100.0 cm^3 solution = 0.0150 mol

$$\text{Molarity of barium hydroxide solution} = \frac{\text{number of moles of } Ba(OH)_2}{\text{volume of solution}}$$

$$= \frac{0.0150\ mol}{\left(\frac{100.0}{1\,000}\right) dm^3}$$

$$= 0.150\ mol\ dm^{-3}$$

$\therefore$ the concentration of the barium hydroxide solution is 0.150 mol dm^{-3}.

Practice 17.5 Refer to page T40.

1 Solid acid T has a relative molecular mass of 192.0. A sample of 0.960 g of T is dissolved in water to form a solution which requires 25.0 cm^3 of 0.400 mol dm^{-3} sodium hydroxide solution for complete neutralization. What is the basicity of T?

2

CE 2010 Paper 2 Q35

2 10.3 g of a sample of hydrated sodium carbonate ($Na_2CO_3 \bullet nH_2O$) were dissolved in water. The solution was made up to 250.0 cm^3. 25.0 cm^3 of this solution were titrated against 0.120 mol dm^{-3} sulphuric acid, using methyl orange as the indicator. 30.0 cm^3 of the sulphuric acid were required to reach the end point.

a) 25.0 cm^3 of the sodium carbonate solution were transferred to a clean conical flask. Briefly describe how the titration of this solution should be carried out.

b) State the colour change of the indicator at the end point of the titration.

c) Calculate *n*, the number of molecules of water of crystallization which combines with one formula unit of sodium carbonate.

10

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0, Na = 23.0)

17.10 Back titration

In an acid-alkali titration, we can determine the concentration (or amount) of a substance in a sample by gradual addition of a standard acid / alkali until complete neutralization occurs. This process is known as direct titration.

However, we may need to modify the procedure in some cases. Suppose we need to determine the mass of aluminium hydroxide in a drug tablet. Aluminium hydroxide is insoluble in water and is thus difficult to titrate directly. We can determine its mass in a drug tablet in the following way:

1 Dissolve one drug tablet in a known volume of a standard acid (in excess). The acid will react with all the aluminium hydroxide in the tablet.

2 Find the amount of acid left over (i.e. the acid unreacted) by titrating it against a standard alkali.

This method is known as **back titration**. The acid used up by the aluminium hydroxide is the acid originally added minus the acid left over at the end. We can calculate the mass of aluminium hydroxide in the drug tablet from the acid used up. Fig. 17.20 summarizes the steps to calculate the mass of aluminium hydroxide in the drug tablet by back titration.

Assessment Form
Assessment forms on these activities are available for download from *Jing Kung Chemistry Website*.

17.5 **Assessment Form**
Determining the mass of the active ingredient in an antacid tablet.

17.6 **Assessment Form**
Determining the percentage by mass of calcium carbonate in eggshells by back titration. N13

N13 Refer to page T42.

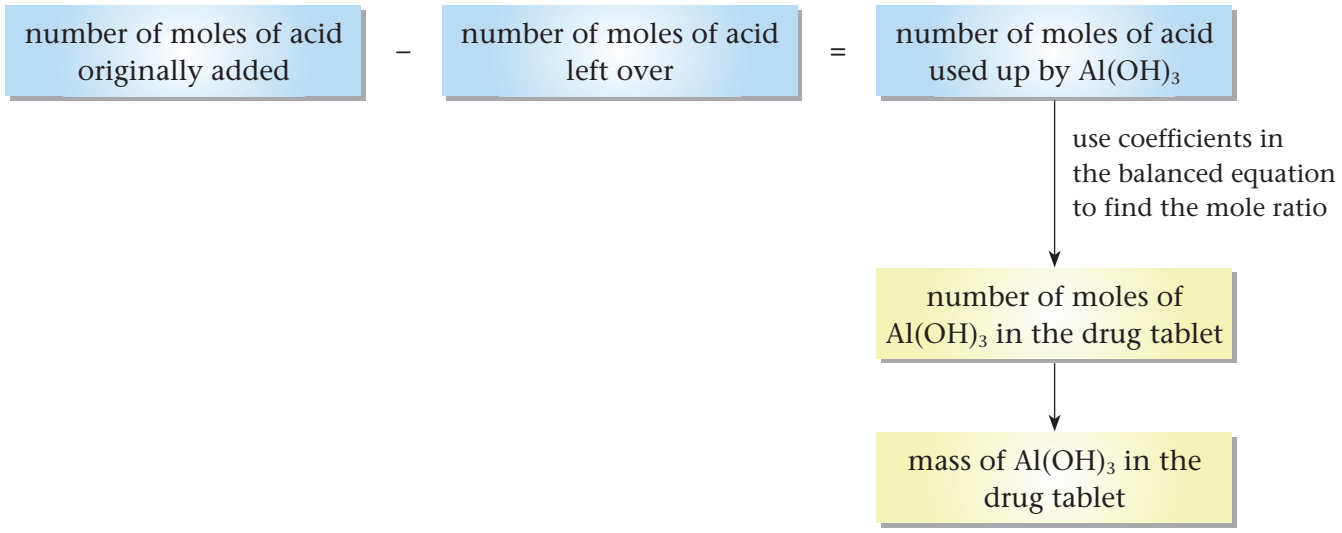
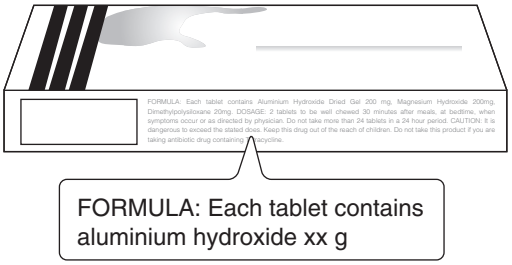


Fig. 17.20 A diagram summarizing the steps to calculate the mass of aluminium hydroxide in a drug tablet by back titration

back titration 返滴定法

Example 17.11

Q The drug tablets shown below can relieve stomach aches. It contains aluminium hydroxide, $Al(OH)_3$, as the only active ingredient.



A student performed the following experiment to determine the amount of aluminium hydroxide contained in one drug tablet.

Step 1 One drug tablet was dissolved in 50.0 cm^3 of 1.00 mol dm^{-3} hydrochloric acid to form a solution.

Step 2 The solution was placed in a volumetric flask and then diluted to 250.0 cm^3 with distilled water.

Step 3 25.0 cm^3 of the diluted solution were titrated with 0.190 mol dm^{-3} sodium hydroxide solution using a suitable indicator. 18.1 cm^3 of sodium hydroxide solution were needed to reach the end point.

a) Calculate the number of moles of excess hydrochloric acid in 25.0 cm^3 of the diluted solution from the data obtained in the titration.

b) Calculate the number of moles of hydrochloric acid that was added in *Step 1*.

c) Hence calculate the mass of aluminium hydroxide in one drug tablet.

(Relative atomic masses: H = 1.0, O = 16.0, Al = 27.0)

CE 2009 Paper 1 Q11

A a) $NaOH(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l)$

0.190 mol dm^{-3}

18.1 cm^3

Number of moles of NaOH in 18.1 cm^3 solution = molarity of solution x volume of solution

$$= 0.190 \text{ mol dm}^{-3} \times \frac{18.1}{1\,000} \text{ dm}^3$$

$$= 0.00344 \text{ mol}$$

According to the equation, 1 mole of HCl requires 1 mole of NaOH for complete neutralization.

i.e. number of moles of excess HCl in 25.0 cm^3 diluted solution = 0.00344 mol

b) Number of moles of HCl added in *Step 1* = molarity of solution x volume of solution

$$= 1.00 \text{ mol dm}^{-3} \times \frac{50.0}{1\,000} \text{ dm}^3$$

$$= 0.0500 \text{ mol}$$

Continued on next page

c) Number of moles of HCl left over after reaction with drug tablet in *Step 1*

$= 0.00344 \text{ mol} \times \frac{250.0 \text{ cm}^3}{25.0 \text{ cm}^3}$

$= 0.0344 \text{ mol}$

Number of moles of HCl reacted with $Al(OH)_3$ in drug tablet $= (0.0500 - 0.0344) \text{ mol}$
$= 0.0156 \text{ mol}$

$Al(OH)_3(s) + 3HCl(aq) \longrightarrow AlCl_3(aq) + 3H_2O(l)$

According to the equation, 1 mole of $Al(OH)_3$ requires 3 moles of HCl for complete neutralization.

i.e. number of moles of $Al(OH)_3$ in drug tablet $= \frac{0.0156}{3} \text{ mol}$
$= 0.00520 \text{ mol}$

Mass of $Al(OH)_3$ in drug tablet = number of moles × molar mass
$= 0.00520 \text{ mol} \times 78.0 \text{ g mol}^{-1}$
$= 0.406 \text{ g}$

$\therefore$ the drug tablet contains 0.406 g of aluminium hydroxide.

Practice 17.6 Refer to page T42.

The following experiment was carried out to determine the concentration of copper(II) ions in a copper(II) nitrate solution:

Step 1 25.0 cm^3 of 0.660 mol dm^{-3} sodium hydroxide solution were added to 25.0 cm^3 of copper(II) nitrate solution to precipitate out the copper(II) hydroxide.

Step 2 The copper(II) hydroxide was removed from the reaction mixture.

Step 3 The alkali left over in the reaction mixture was titrated against 0.200 mol dm^{-3} hydrochloric acid using a suitable indicator. 20.5 cm^3 of the acid were required to reach the end point.

a) Write an ionic equation for the reaction that occurred in *Step 1*.
$Cu^{2+}(aq) + 2OH^-(aq) \longrightarrow Cu(OH)_2(s)$

b) Suggest a method to remove the copper(II) hydroxide from the reaction mixture in *Step 2*.
By filtration

c) i) Based on the titration result in *Step 3*, calculate the number of moles of hydroxide ions left over in the reaction mixture.
0.00410 mol

ii) Calculate the number of moles of sodium hydroxide that was added in *Step 1*.
0.0165 mol

iii) Calculate the concentration of copper(II) ions in the copper(II) nitrate solution.
0.248 mol dm^{-3}

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

Chemistry Magazine

Sulphur dioxide content in wine

The main compound in wine (Fig. 17.21) is ethanol (CH_3CH_2OH). The oxygen dissolved in the wine may oxidize ethanol to ethanoic acid (CH_3COOH) and turn wine into vinegar. Sulphur dioxide is used as a preservative to prevent the oxidation of ethanol.



Fig. 17.21 Sulphur dioxide is added to this bottle of white wine to prevent the oxidation of ethanol

Determining the sulphur dioxide content in wine

The aeration oxidation method is commonly used to determine the sulphur dioxide content in wine.

A wine sample is allowed to react with an acid in a distillation flask. The sulphur dioxide bubbles out of the solution as a gas. The gas then passes into a hydrogen peroxide (H_2O_2) solution. The sulphur dioxide reacts with the peroxide to form sulphuric acid according to the following equation:

$$SO_2(g) + H_2O_2(aq) \longrightarrow H_2SO_4(aq)$$

The sulphuric acid formed is then titrated against a standard sodium hydroxide solution. The amount of sulphuric acid and hence the sulphur dioxide content in the wine sample can be determined.

Questions

1 Volumetric analysis is widely used in food analysis. Suggest examples based on what you have learnt in this topic.

2 Sulphur dioxide is a preservative — a food additive. Food additives are added to foods to make them last longer, look more attractive or improve in flavours.

a) State other classes of food additives besides preservatives.

b) Discuss the advantages and disadvantages of using food additives.

Refer to page T43.

aeration oxidation method 充氣氧化法

Have you mastered?

Key terms

concentration 濃度 91 dilution 稀釋 94 volumetric analysis 容量分析 95

electronic balance 電子天平 96 volumetric flask 容量瓶 96 pipette 移液管 96

pipette filler 移液管膠泵 96 burette 滴定管 96 standard solution 標準溶液 97

ethanedioic acid crystals 乙二酸晶體 97 weighing bottle 稱量瓶 98 meniscus 彎液面 98

titration 滴定 100 acid-alkali indicator 酸鹼指示劑 101 titre 滴定值 102

equivalence point 當量點 105 titration curve 滴定曲線 105 titration end point 滴定終點 106

methyl orange 甲基橙試液 107 phenolphthalein 酚酞試液 107 back titration 返適定法 118

Checklist

After studying this unit, you should be able to

- ☐ describe how to prepare a solution of the required concentration by dissolving a solid or diluting a concentrated solution;
- ☐ calculate the concentration of the solution prepared;
- ☐ describe the techniques of performing acid-alkali titrations;
- ☐ use the results of acid-alkali titrations to solve numerical problems.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 Concentration of a solution ($g\ dm^{-3}$) = $\dfrac{\text{mass of solute (g)}}{\text{volume of solution } (dm^3)}$

2 The numbers of moles of solute in a solution before dilution and after dilution are the same, i.e.

(MV) before dilution = (MV) after dilution, where M = molarity, V = volume.

3 In volumetric analysis, just sufficient volume of a solution of known concentration is allowed to completely react with the substance being analyzed in a sample. The process of determining the 'just sufficient' volume is called titration. From the measured volume and known concentration of the solution used, we can calculate the concentration (or amount) of the substance in question.

4 A standard solution is a solution with accurately known concentration.

5 The point in an acid-alkali titration at which the acid and alkali just react completely with each other is called the equivalence point.

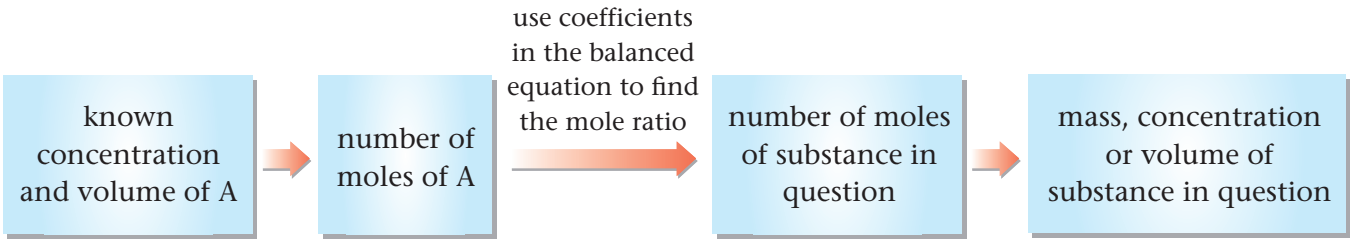
6 An acid-alkali indicator is used to indicate the end point of an acid-alkali titration.

Type of acid-alkali titration	Suitable indicator
Strong acid-strong alkali	methyl orange or phenolphthalein
Weak acid-strong alkali	phenolphthalein
Strong acid-weak alkali	methyl orange
Weak acid-weak alkali	—

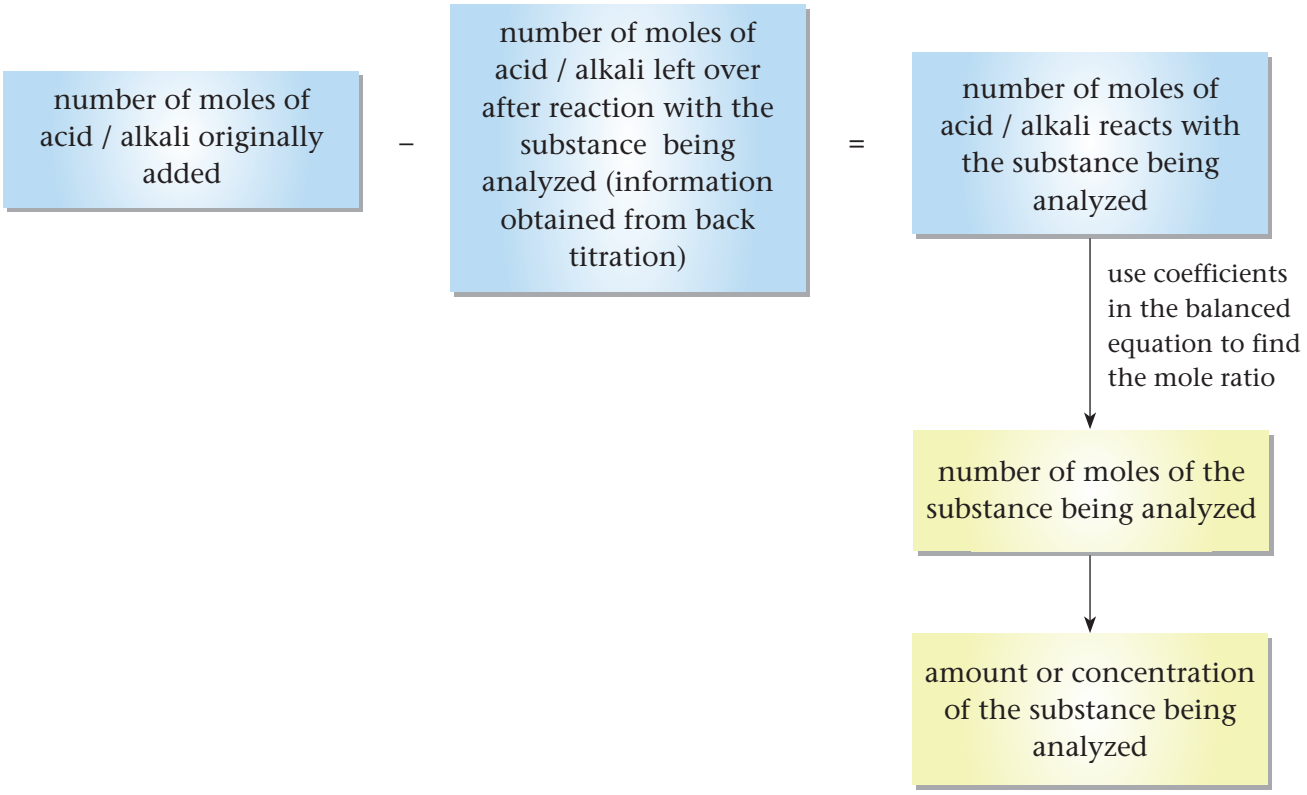
7 Special apparatus used in volumetric analysis include

- electronic balance;
- volumetric flask;
- pipette and pipette filler;
- burette.

8 The following diagram summarizes the steps for calculations in volumetric analysis.



9 The following diagram summarizes the steps to calculate the amount or concentration of the substance being analyzed in a sample by back titration.



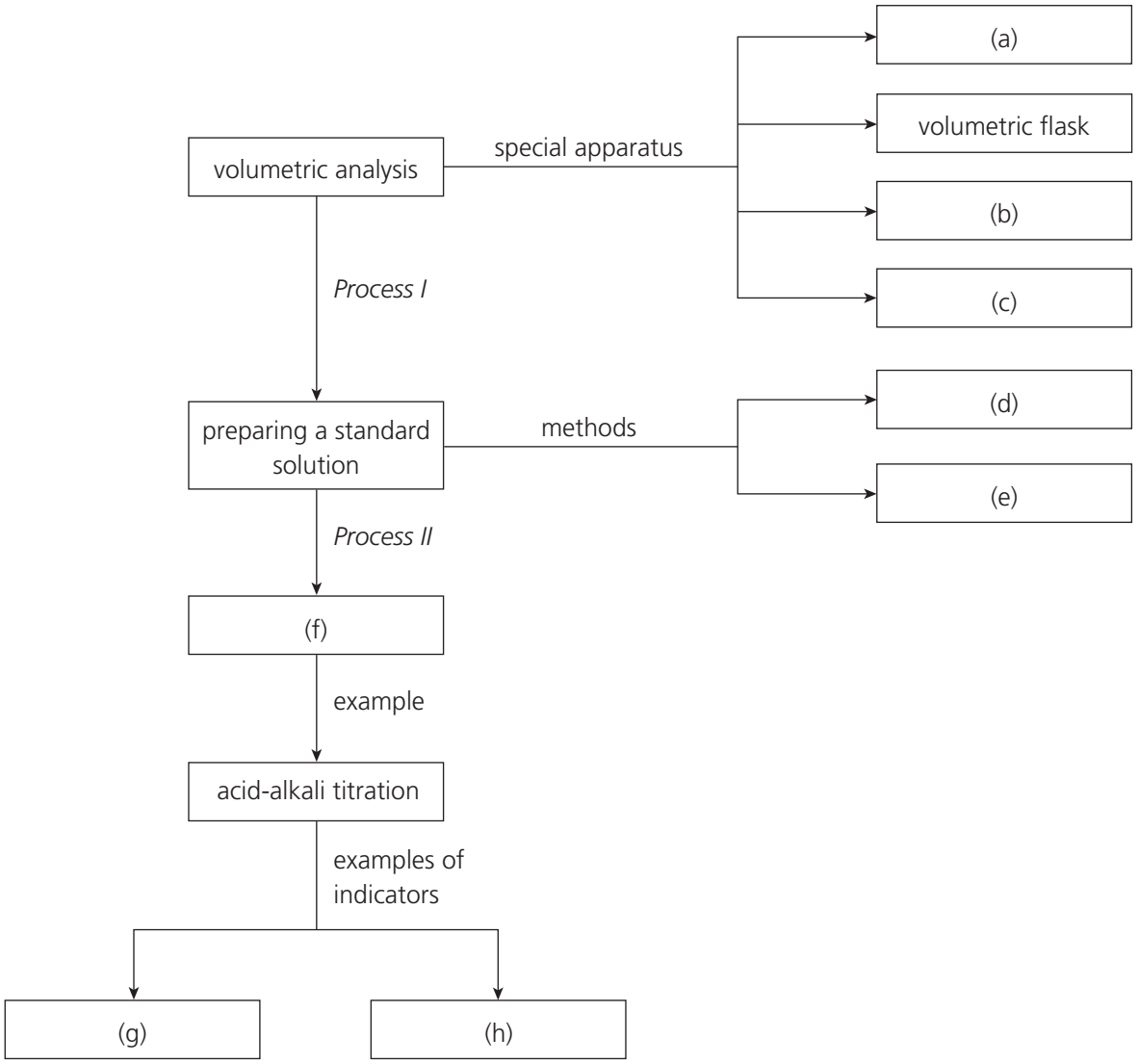
Have you mastered?

Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

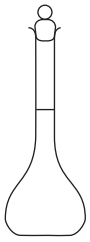
Part I Knowledge and understanding

1 Complete the following concept map.



a) electronic balance
b) pipette and pipette filler
c) burette
d) dissolving a solid acid / alkali in water
e) diluting a concentrated acid / alkali of known concentration
f) titration
g) methyl orange
h) phenolphthalein

2 Name the following pieces of apparatus.

(a) 

(b) 

(c) 

(d) 

a) Volumetric flask
b) Pipette
c) Conical flask
d) Burette

Part II Multiple choice questions

3 A sample of concentrated sulphuric acid has a density of 1.93 g cm^{-3} and contains 93.5% of sulphuric acid by mass. What is the concentration of the sulphuric acid sample?

Refer to page T44 for explanation.

(Relative atomic masses: H = 1.0, O = 16.0, S = 32.1)

A 17.6 mol dm^{-3}
(B) 18.4 mol dm^{-3}
C 19.8 mol dm^{-3}
D 21.0 mol dm^{-3}

4 What volume of water is required to dilute 30.0 cm^3 of 8.25 mol dm^{-3} hydrochloric acid to a concentration of 2.50 mol dm^{-3}?

Refer to page T45 for explanation.

A 30.0 cm^3
(B) 69.0 cm^3
C 99.0 cm^3
D 120.0 cm^3

5 1.24 dm^3 of 5.00 mol dm^{-3} potassium hydroxide solution are diluted to 20.0 dm^3. What is the concentration of the diluted solution in g dm^{-3}?

(Relative atomic masses: H = 1.0, O = 16.0, K = 39.1)

Refer to page T45 for explanation.

A 0.310 g dm^{-3}
B 8.06 g dm^{-3}
C 12.5 g dm^{-3}
(D) 17.4 g dm^{-3}

6 Besides a pipette, which of the following apparatus must be used in order to prepare 250.0 cm^3 of 0.100 M Na_2CO_3(aq) from 1.00 M Na_2CO_3(aq)?

A Burette
B Conical flask
(C) Volumetric flask
D Measuring cylinder

(HKCEE, Paper 2, 2010, 19)

7 10.0 cm^3 of an aqueous solution of a dibasic acid require 25.0 cm^3 of 0.240 mol dm^{-3} NaOH(aq) for complete neutralization. What is the concentration of the acid solution?

Refer to page T45 for explanation.

(A) 0.300 mol dm^{-3}
B 0.600 mol dm^{-3}
C 0.900 mol dm^{-3}
D 1.20 mol dm^{-3}

8 30.0 cm^3 of 0.10 M KOH are completely neutralized by 20.0 cm^3 of dilute H_2SO_4 to form K_2SO_4 solution. What is the molarity of the salt solution obtained?

(A) 0.03 M
B 0.05 M
C 0.06 M
D 0.10 M

(HKCEE, Paper 2, 2008, 7)

9 Solid acid T has a relative molecular mass of 192.0. A sample of 0.80 g of T is dissolved in water to form a solution which requires 25.0 cm^3 of 0.50 M sodium hydroxide solution for complete neutralization. What is the basicity of T?

A 1
B 2
C 3
D 4

(HKCEE, Paper 2, 2010, 35)

10 In a titration experiment, which of the following apparatus should be rinsed with the solution it is about to contain?

(1) Burette
(2) Conical flask
(3) Volumetric flask

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

11 Solution Y is added dropwise to a solution of NaOH containing several drops of phenolphthalein. The mixture suddenly changes from pink to colourless. Which of the following substances may Y be?

(1) HCl(aq)
(2) KCl(aq)
(3) Cl_2(aq)

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2010, 43)

12 In an experiment, 10 cm^3 of 1.0 M sulphuric acid are mixed with 30 cm^3 of 0.5 M sodium hydroxide solution. Which of the following statements concerning this experiment is / are correct?

(1) 0.015 mole of water is formed.
(2) The pH of the resulting mixture is greater than 7.
(3) After water is completely evaporated from the resulting mixture, pure sodium sulphate solid can be obtained.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2011, 43)

Part III Structured questions

Answers for the questions in Hong Kong public examinations are not provided.

13 This question is about the pH of several solutions.
Refer to page T46.

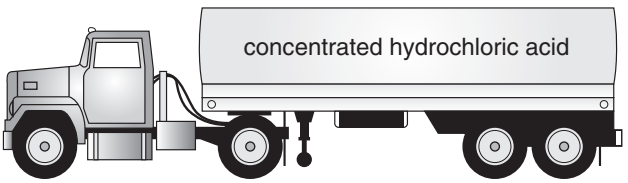
a) Write an expression for pH.
$pH = -\log_{10}[H^+]$

b) Calculate the pH of 0.154 mol dm^{-3} hydrochloric acid.
0.812

b) Calculate the pH of the solution formed when 10.0 cm^3 of 0.154 mol dm^{-3} hydrochloric acid are added to 990.0 cm^3 of water.
2.81

(AQA Advanced GCE, Unit 4, Jan. 2011, 2(a))

14 The diagram below shows a truck with a storage tank for transporting concentrated hydrochloric acid.



a) Suggest a hazard warning label that should be posted on the storage tank.

b) The storage tank contains 57 000 kg of concentrated hydrochloric acid, which occupies a volume of 50 m^3. If the percentage by mass of HCl in the acid is 38.0%, calculate the molarity of the acid.

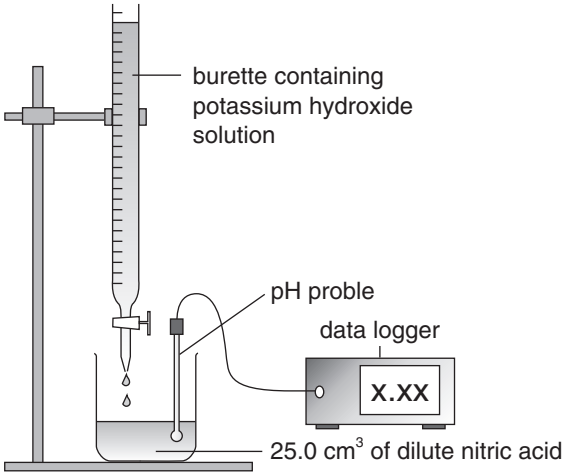
(HKCEE, Paper 1, 2009, 4(a), (c))

15 Zak investigates the neutralization of dilute nitric acid.

He measures out 25.0 cm^3 of 0.150 mol dm^{-3} dilute nitric acid. He puts this acid into a beaker.

Zak reacts the dilute nitric acid with an alkali, potassium hydroxide solution.

Look at the diagram. It shows the apparatus he uses.

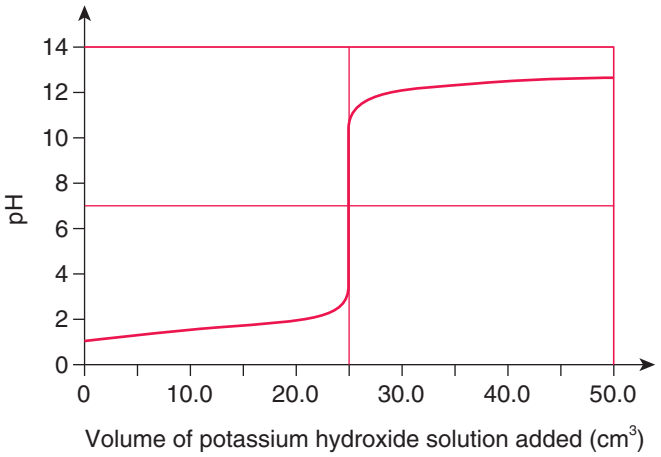


Zak slowly adds the alkali to the dilute nitric acid.

He uses a pH probe to measure the pH of the solution in the beaker.

a) The pH of the solution in the beaker changes as more and more potassium hydroxide solution is added.

Sketch a graph to show how the pH changes.



Curve starting from pH below 7 and finishing at pH above 7.
Curve includes an almost vertical section centred on pH 7.

b) Zak uses 25.0 cm^3 of 0.150 mol dm^{-3} dilute nitric acid.

0.00375 mol; refer to page T46.

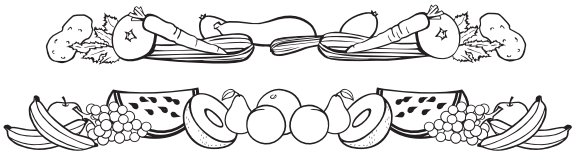
How many moles of nitric acid does Zak use?

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 2, Jan. 2011, 8(a)–(b))

16 Ammonium sulphate is an artificial fertilizer.

Refer to page T46.

Feed the World



Fertilizer

Contains

Ammonium sulphate $(NH_4)_2SO_4$

a) When this fertilizer is warmed with sodium hydroxide solution, ammonia gas is given off. Describe and give the result of a test for ammonia gas.

b) i) A student made some ammonium sulphate in a school laboratory.

The student carried out a titration, using a suitable indicator, to find the volumes of sulphuric acid and aqueous ammonia that should be reacted together.

Name a suitable indicator for strong acid-weak alkali titrations.

ii) The student found that 25.0 cm^3 of aqueous ammonia reacted completely with 32.0 cm^3 of sulphuric acid of concentration 0.0500 mol dm^{-3}.

The equation that represents this reaction is:

$$H_2SO_4(aq) + 2NH_3(aq) \longrightarrow (NH_4)_2SO_4(aq)$$

Calculate the concentration of this aqueous ammonia in mol dm^{-3}.

0.128 mol dm^{-3}

iii) Use your answer to (ii) to calculate the concentration of ammonia in g dm^{-3}.

2.18 g dm^{-3}

(Relative molecular mass of ammonia (NH_3) = 17.0)

(AQA GCSE (Higher Tier), Chemistry, Unit 3, May 2011, 5)

17 The concentration of a sodium hydroxide solution can be found by titration.



The photograph shows a student preparing to carry out the experiment.

$NaOH(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l)$

The student titrated 10.00 cm^3 portions of the sodium hydroxide solution with 0.100 mol dm^{-3} hydrochloric acid, HCl, using methyl orange as the indicator. One trial titration and three further titrations were carried out.

The results of the titrations are

volume of sodium hydroxide solution used = 10.00 cm^3

volume of 0.100 mol dm^{-3} hydrochloric acid needed
trial titration = 10.40 cm^3
2nd titration = 9.80 cm^3
3rd titration = 10.10 cm^3
4th titration = 9.70 cm^3

a) After the trial titration, why were more titrations carried out?

b) Why is an indicator used in the titration?

c) What volume of hydrochloric acid should be used in the calculation to find the concentration of the sodium hydroxide solution?

d) Describe how the student should carry out the titration.

e) A different sodium hydroxide solution, NaOH, has a concentration of 0.080 mol dm^{-3}. Calculate the concentration of this solution in g dm^{-3}.

(Formula mass of NaOH = 40.0)

(Edexcel GCSE, Chemistry, Unit C3, May 2012, 9)

18 Gemma works for a company making vinegar.

She measures the amount of ethanoic acid in 25.0 cm^3 samples of the company's product.

She carries out a titration using a standard solution of sodium hydroxide and an indicator. Gemma carries out six titrations in the morning and six more in the afternoon.

All of the samples she tests are from the same vinegar.

Her results are shown in the table.

	Volume of sodium hydroxide solution (cm^3)					
Morning	12.9	12.2	12.5	12.8	12.9	12.1
Afternoon	12.4	12.6	12.5	12.5	12.4	12.6

Gemma decides to use the results she obtained in the afternoon to calculate the concentration of ethanoic acid in the vinegar.

a) Explain why she chose the afternoon set of results.
The results are more consistent.

b) Ethanoic acid and sodium hydroxide solution react according to this equation.

$CH_3COOH(aq) + NaOH(aq) \longrightarrow CH_3COONa(aq) + H_2O(l)$

Gemma used 25.0 cm^3 of vinegar for each titration. The average of the results from Gemma's afternoon titrations is 12.5 cm^3.

Use this average, and the concentration of sodium hydroxide solution (0.100 mol dm^{-3}), to calculate the mass of ethanoic acid in each dm^3 of the vinegar.
3.00 g; refer to page T47.
(Relative molecular mass: CH_3COOH = 60.0)

(OCR GCSE 21st Century Science (Higher Tier), Chem. A, Unit 3, Jun. 2008, 4(b))

17 a) To obtain consistent results.
b) To show the end point of the reaction.
c) $\frac{9.70 + 9.80}{2}$ cm^3 = 9.75 cm^3
d)
- Wash the burette with acid. / Wash the pipette with alkali.
- Deliver 10.0 cm^3 of the alkali to a conical flask using the pipette. / Fill the burette with the acid.
- Add a few drops of methyl orange to the alkali.
- Run the acid from the burette into the alkali and swirl.
- Add the acid until the indicator turns orange.

e) 0.080 mol dm^{-3} x 40.0 g mol^{-1} = 3.2 g dm^{-3}

19 Snottites are slimy colonies of bacteria that live in extreme conditions in caves. They take in hydrogen sulphide gas, H_2S, and oxygen gas and form sulphuric acid, H_2SO_4, in their slime.

A sample of the sulphuric acid formed by snottites was collected in order to find its concentration.

A scientist measured out a 10.0 cm^3 sample of cave water containing sulphuric acid using a pipette and then diluted it to 250.0 cm^3 in a volumetric flask.

The scientist then carried out a titration of the diluted acid with sodium hydroxide solution.

In the titration, 20.0 cm^3 of the dilute sulphuric acid solution were found to react exactly with 26.40 cm^3 of 0.0500 mol dm^{-3} sodium hydroxide solution.

$$H_2SO_4(aq) + 2NaOH(aq) \longrightarrow Na_2SO_4(aq) + 2H_2O(l)$$

a) Name the piece of apparatus that the scientist would use to add the sodium hydroxide solution to the sulphuric acid during the titration.
Burette

b) Calculate the concentration of the sulphuric acid in the cave water.
0.825 mol dm^{-3}; refer to page T47.

(OCR Advanced Subsidiary GCE, Chem. B (Salters), F332, Jan. 2012, 1(a)(i), (v))

20 Quicklime, CaO, and slaked lime, $Ca(OH)_2$, are bases made by the chemical industry with uses in building and farming.
Refer to page T48.

a) i) How is quicklime made by the chemical industry?
Thermal decomposition of limestone

ii) Give ONE use of slaked lime in farming.
Reduce the acidity of soil.

b) Slaked lime is slightly soluble in water forming a solution commonly referred to as limewater. A student carried out a titration to find the concentration of limewater. 25.0 cm^3 of limewater were neutralized by 22.45 cm^3 of 0.0105 mol dm^{-3} nitric acid, HNO_3.

i) Balance the equation for the reaction that takes place.

$$_1_Ca(OH)_2(aq) + _2_HNO_3(aq) \longrightarrow _1_Ca(NO_3)_2(aq) + _2_H_2O(l)$$

ii) Calculate how many moles of HNO_3 were used.
2.36×10^{-4} mol

iii) Determine how many moles of $Ca(OH)_2$ reacted with the HNO_3.
1.18×10^{-4} mol

iv) Calculate the concentration, in mol dm^{-3}, of the $Ca(OH)_2$ in the limewater.
0.00472 mol dm^{-3}

v) After the titration, the student allowed the water to evaporate to obtain a hydrated crystalline solid with a molar mass of 272.1 g mol^{-1}.

Determine the formula of the hydrated solid.
$Ca(NO_3)_2 \bullet 6H_2O$
(Relative atomic masses: H = 1.0, N = 14.0, O = 16.0, Ca = 40.1)

(OCR Advanced Subsidiary GCE, Chemistry, 2811/01, Jan. 2008, 2)

21 A student was given a sample of a water-soluble metal carbonate, $M_2CO_3(s)$. In order to deduce what M was, the student prepared a 100.0 cm^3 aqueous solution of the carbonate using 1.14 g of the sample. The student then withdrew several 10.0 cm^3 portions of the solution, and titrated each portion with 0.0850 mol dm^{-3} HCl(aq) using methyl orange as indicator. The mean titre was 25.30 cm^3.

a) Describe how the 100.0 cm^3 aqueous solution was prepared.

b) Based on the experimental results, determine the formula mass of M_2CO_3 and deduce what M is.

(Relative atomic masses: C = 12.0, O = 16.0)

(HKDSE, Practice paper 1B, 2012, 4)

Unit Exercise

22 Lime scale can build up in kettles.
Refer to page T49.
A company makes a lime scale remover that contains solid phosphoric acid, H_3PO_4, together with other ingredients. When dissolved in water the phosphoric acid reacts with and removes the lime scale.

The batch of lime scale remover that has been made each day is analyzed to measure how much phosphoric acid it contains.

a) The amount of phosphoric acid in 10.0 g samples of the lime scale remover is measured using titration against standard sodium hydroxide solution. A rough titration is carried out first, then several accurate titrations.

Describe the key stages of an accurate titration.

b) The titration is repeated with several other samples of the lime scale remover.

Give TWO reasons for analyzing several samples of the lime scale remover.

c) Why is it important for the company to know how much phosphoric acid the lime scale remover contains?

d) The result of the analysis shows that 25.0 cm^3 of standard sodium hydroxide solution react with the phosphoric acid in 10.0 g of the lime scale remover.

i) The standard sodium hydroxide solution contains 60.0 g dm^{-3} of sodium hydroxide. What is the mass of sodium hydroxide in 25.0 cm^3 of this standard solution?
1.50 g

ii) Phosphoric acid reacts with sodium hydroxide solution according to this equation.

$$H_3PO_4(aq) + 3NaOH(aq) \longrightarrow Na_3PO_4(aq) + 3H_2O(l)$$

Work out the mass of phosphoric acid in 10.0 g of lime scale remover.
1.23 g
(Relative atomic masses: H = 1.0, O = 16.0, Na = 23.0, P = 31.0)

(OCR GCSE 21st Century Science (Higher Tier), Chem. A, Unit 3, Jun. 2011, 4)

23 Chemicals called 'acids' have been known throughout history. The word acid comes from the Latin 'acidus' meaning sour. Dilute sulphuric acid, H_2SO_4, is a common laboratory acid.
Refer to page T50.

a) i) State the formulae of two ions released when sulphuric acid is in aqueous solution.

ii) A student adds a sample of solid potassium carbonate, K_2CO_3, to an excess of dilute sulphuric acid.

Describe what the student would see and write the equation for the reaction which takes place.

b) Dilute sulphuric acid reacts with alkalis such as sodium hydroxide.

Solid sodium hydroxide is known as caustic soda. It has a household use as a drain cleanser.

A student believes a box of caustic soda has been accidentally contaminated.

- To prove this, the student dissolves 2.00 g of the impure caustic soda in water and the solution is made up to 250.0 cm^3.
- 25.0 cm^3 of this solution of caustic soda are neutralized by 24.60 cm^3 of 0.100 mol dm^{-3} dilute sulphuric acid.

$$H_2SO_4(aq) + 2NaOH(aq) \longrightarrow Na_2SO_4(aq) + 2H_2O(l)$$

Calculate the percentage, by mass, of NaOH in the impure caustic soda. 98.5%

(OCR Advanced Subsidiary GCE, Chem. A, F321, Jun. 2010, 2(a), (b)(iii))

24 In 1916, during the First World War, a German U-boat sank a Swedish ship which was carrying a cargo of champagne. The wreck was discovered in 1997 and the champagne was brought to the surface and analyzed.

Refer to the next page.

a) 25.0 cm^3 of the champagne were placed in a conical flask.

Describe how the volume of sodium hydroxide solution needed to react completely with the weak acids in 25.0 cm^3 of this champagne can be found by titration, using phenolphthalein indicator.

Name any other apparatus used.

b) The acid in 25.0 cm^3 of the champagne reacted completely with 13.5 cm^3 of sodium hydroxide solution of concentration 0.100 mol dm^{-3}.

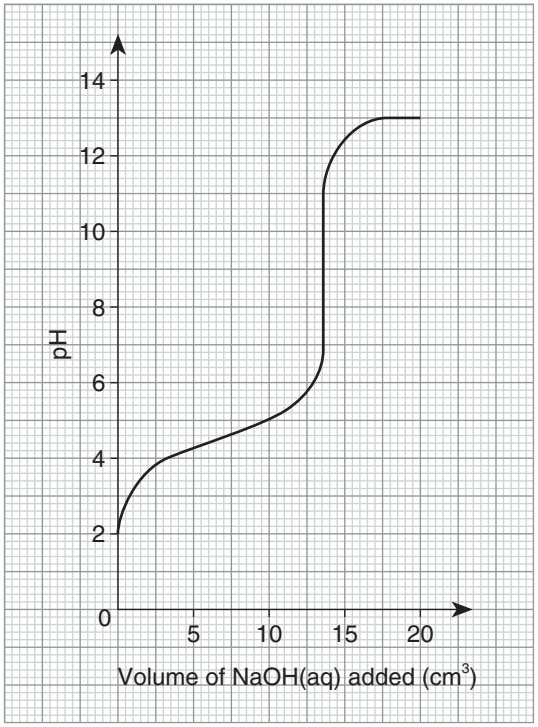
Calculate the concentration in mol dm^{-3} of acid in the champagne.

Assume that 1 mole of sodium hydroxide reacts completely with 1 mole of acid.

c) Is analysis by titration enough to decide whether this champagne is safe to drink?

Explain your answer.

d) The graph shows how the pH of the solution changes during this titration.



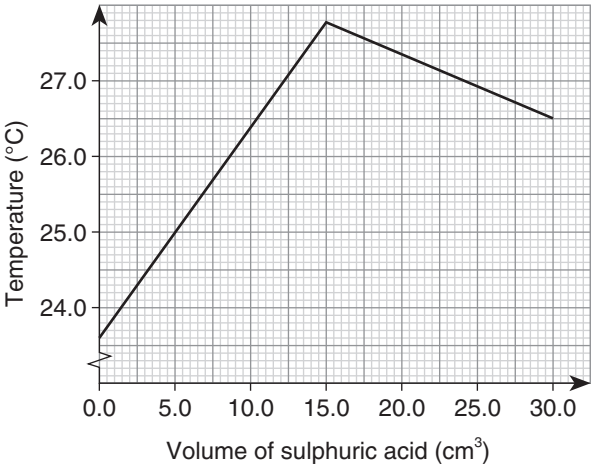
Phenolphthalein is the indicator used in this titration. It changes colour between pH 8.2 and pH 10.0.

Methyl orange is another indicator. It changes colour between pH 3.2 and pH 4.4.

Suggest why methyl orange is NOT a suitable indicator for this titration.

(AQA GCSE (Higher Tier), Chemistry, Unit 3, Jan. 2008, 5)

25 In an experiment, 25.00 cm^3 of sodium hydroxide solution are transferred to an expanded polystyrene cup. 0.50 M sulphuric acid is then added to the solution from a burette, and the temperature of the mixture is measured with a data-logger. The graph below shows the experimental results:



a) Name the apparatus that should be used to transfer 25.00 cm^3 of sodium hydroxide solution to the expanded polystyrene cup.

b) Outline the procedure for cleaning the burette before experiment.

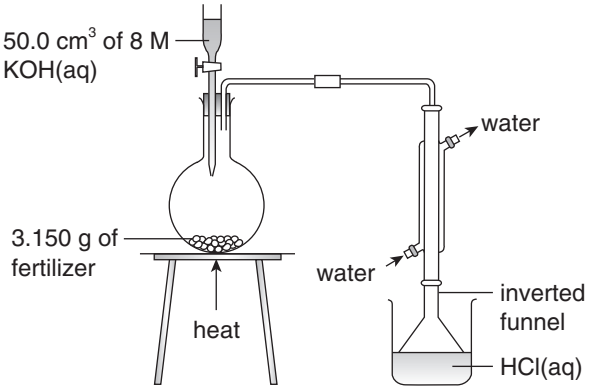
c) Write an ionic equation for the reaction involved.

d) With reference to the above graph, explain the temperature change of the mixture throughout the experiment.

e) Calculate the molarity of the sodium hydroxide solution used.

(HKCEE, Paper 1, 2010, 10)

26 A fertilizer only contains ammonium nitrate (NH_4NO_3) and potassium chloride (KCl). An experiment was performed to determine the percentage by mass of NH_4NO_3 in this fertilizer. The set-up used is shown below:



The KOH(aq) was added slowly to the fertilizer and the mixture formed was heated gently. The ammonia liberated from the reaction between NH_4NO_3 and KOH was first cooled in a condenser, and then passed through an inverted funnel to a solution containing 0.0485 mol of HCl. The solution was finally made up to 100.0 cm^3 and labelled 'S'.

a) Write an ionic equation for the reaction between NH_4NO_3 and KOH.

b) Suggest the potential hazard of one of the chemicals used.

c) Given that ammonia is very soluble in water, state the advantage of using an inverted funnel.

d) 25.00 cm^3 of 'S' were transferred to a conical flask, and then titrated with 0.100 M NaOH(aq) using methyl orange as an indicator. 41.00 cm^3 of the NaOH(aq) were required to reach the end point.

i) Name the apparatus that should be used to transfer 25.00 cm^3 of 'S'.

ii) State the colour change at the end point of the titration.

iii) Calculate the percentage by mass of NH_4NO_3 in this fertilizer.

(Molar mass of NH_4NO_3 = 80.0 g mol^{-1})

e) Suggest a test to show the presence of a potassium-containing compound in the fertilizer.

(HKDSE, Paper 1B, 2012, 7)

27 In an experiment to determine the relative atomic mass of magnesium, 0.420 g of magnesium ribbon was added to 25.0 cm^3 of 0.955 mol dm^{-3} H_2SO_4(aq). When effervescence ceased, the resulting mixture was diluted to 250.0 cm^3 with deionized water. 25.0 cm^3 portions of the diluted solution were withdrawn and titrated against 0.0941 mol dm^{-3} NaOH(aq) using methyl orange as indicator. The mean titre was 16.48 cm^3.

a) State the colour change at the end point of the titration.

b) Based on the titration results, calculate the relative atomic mass of magnesium.

c) Assuming that the experimental error is negligible, suggest ONE reason why the relative atomic mass of magnesium calculated in (b) is different from that found in the Periodic Table.

(HKASLE, Paper 2, 2009, 3)

24 a)
- Add a few drops of phenolphthalein to the champagne.
- Fill a burette with the standard sodium hydroxide solution. Record the initial burette reading.
- Run the standard sodium hydroxide solution into the champagne until the reaction mixture changes from colourless to pink. Record the final burette reading.
- Repeat the titration to obtain three consistent titres (titres that are within 0.1 cm^3).

b) Number of moles of NaOH in 13.5 cm^3 solution
= molarity of solution x volume of solution
$= 0.100 \text{ mol dm}^{-3} \times \frac{13.5}{1\,000} \text{ dm}^3$
= 0.00135 mol
1 mole of NaOH reacts completely with 1 mole of acid.
i.e. number of moles of acid in 25.0 cm^3 of champagne
= 0.00135 mol
Concentration of acid in champagne
$= \frac{0.00135 \text{ mol}}{\left(\frac{25.0}{1\,000}\right) \text{dm}^3}$
= 0.0540 mol dm^{-3}
∴ the concentration of acid in the champagne is 0.0540 mol dm^{-3}.

c) Any one of the following:
- Yes — the acid concentration is low compared with the acid secreted by the stomach.
- No — acid level too high / other substances or bacteria may be present.
- Cannot decide — insufficient evidence to decide / depends on whether the acid level is considered safe or not.

d) Methyl orange changes colour before the pH range of the vertical part of the curve.

Topic Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Multiple choice questions

1 Which of the following pairs of aqueous solutions, when mixed, would give a precipitate?

(A) Lead(II) nitrate and ammonia
B Copper(II) sulphate and sodium nitrate
C Calcium chloride and sodium nitrate
D Iron(II) sulphate and acidified potassium dichromate

(HKDSE, Paper 1A, 2013, 11)

!!2 Which of the following statements about potassium hydroxide solution is INCORRECT?

(A) When potassium hydroxide solution is added to iron(III) sulphate solution, a dirty green precipitate is formed.
B When potassium hydroxide solution is heated with ammonium chloride solution, ammonia gas is liberated.
C Dilute potassium hydroxide solution contains $K^+(aq)$ ions, $H^+(aq)$ ions and $OH^-(aq)$ ions.
D Concentrated potassium hydroxide solution is corrosive.

(HKDSE, Paper 1A, 2013, 9)

3 Solid Y is soluble in cold water. When an aqueous solution of Y is added separately to sodium hydroxide solution and to acidified silver nitrate solution, a white precipitate is formed in both cases. Which of the following compounds might Y be?

A Ammonium carbonate
B Zinc carbonate
C Lead(II) chloride
(D) Magnesium chloride

(HKDSE, Paper 1A, 2013, 3)

!4 10 cm^3 of 0.25 mol dm^{-3} calcium nitrate solution are mixed with 40 cm^3 of 0.10 mol dm^{-3} nitric acid. What is the concentration of nitrate ions in the resulting solution?

(A) 0.18 mol dm^{-3}
B 0.13 mol dm^{-3}
C 0.080 mol dm^{-3}
D 0.050 mol dm^{-3}

(HKDSE, Practice paper 1A, 2012, 13)

5 Which of the following statements concerning CH_3COOH and HCl is correct?

A CH_3COOH is a stronger acid than HCl.
B The pH of 0.1 M $CH_3COOH(aq)$ is lower than that of 0.1 M HCl(aq).
(C) Both $CH_3COOH(aq)$ and HCl(aq) react with $NH_3(aq)$, each giving a salt.
D Both $CH_3COOH(aq)$ and HCl(aq) react with Ag(s), each giving a colourless gas.

(HKDSE, Paper 1A, 2012, 4)

6 Which of the following gases can be dried by using concentrated sulphuric acid?

(1) Ammonia
(2) Sulphur dioxide
(3) Hydrogen chloride

A (1) only
B (2) only
C (1) and (3) only
(D) (2) and (3) only

(HKCEE, Paper 2, 2011, 20)

Directions : *Each question (Questions 7–10) consists of two separate statements. Decide whether each of the two statements is true or false; if both are true, then decide whether or not the second statement is a correct explanation of the first statement. Then select one option from A to D according to the following table :*

A Both statements are true and the 2nd statement is a correct explanation of the 1st statement.
B Both statements are true but the 2nd statement is NOT a correct explanation of the 1st statement.
C The 1st statement is false but the 2nd statement is true.
D Both statements are false.

	1st statement	2nd statement	
7	Solid citric acid can turn dry blue litmus paper red.	Solid citric acid contains hydrogen ions. *(HKCEE, Paper 2, 2010, 28)*	D
8	Dilute ethanoic acid can conduct electricity.	Ethanoic acid molecules ionize in water to produce mobile ions. *(HKCEE, Paper 2, 2011, 29)*	A
9	If concentrated hydrochloric acid is dripped onto one's hand, one should wash the hand immediately with concentrated ammonia solution.	Concentrated ammonia solution is a weak alkali. *(HKCEE, Paper 2, 2008, 30)*	C
10	All salt solutions are neutral.	All salts are formed from neutralization. *(HKCEE, Paper 2, 2009, 48)*	D

Part II Structured questions

Answers for the questions in Hong Kong public examinations are not provided.

11 In an experiment, carbon dioxide is passed into limewater until excess.

a) State the expected observations and write the chemical equations for the reactions involved.

b) Explain whether the similar observations in (a) would be made if sodium hydroxide solution is used instead of limewater.

c) Explain whether the similar observations in (a) would be made if air is used instead of carbon dioxide.

d) Carbon dioxide can be obtained from the reaction of solid sodium carbonate with dilute hydrochloric acid. Write an ionic equation for the reaction.

(HKCEE, Paper 1, 2010, 6)

12 A solution contains both sodium carbonate and sodium sulphate. Dilute hydrochloric acid, followed by dilute barium chloride solution, is added to this solution to confirm the presence of carbonate ions and sulphate ions.

Refer to page T51.

a) State what would be observed when an excess of dilute hydrochloric acid is added to this mixture. Identify the product responsible for this observation. Write an equation for the reaction which occurs.

b) State what would be observed when an excess of dilute barium chloride solution is added to the solution formed in part (a). Identify the product responsible for this observation. Write an equation for the reaction which occurs.

(AQA Advanced Subsidiary GCE, Chemistry, Unit 1, Jan. 2008, 4)

13 Nicotinic acid is a weak acid. It has the formula C_5H_4NCOOH. Hydrochloric acid is a strong acid. It has the formula HCl.

Refer to page T51.

a) Give the name or formula of the ion that makes solutions acidic.

b) Weak acids contain fewer of these ions than strong acids of the same concentration.

Describe and give the results of an experiment to show that nicotinic acid is a weaker acid than hydrochloric acid of the same concentration.

(AQA GCSE (Higher Tier), Chemistry, Unit C3, Jun. 2008, 1(a))

14 Calcium hydroxide is used in agriculture but the amounts used must be carefully controlled.

Refer to page T51.

a) State ONE use of calcium hydroxide in agriculture and suggest why the amount of calcium hydroxide used should not be excessive.

b) A student knew that calcium hydroxide could be made by adding calcium to water.

The student added 0.00131 mole of calcium to a beaker containing about 100 cm^3 of water.

A reaction took place as shown by the equation below.

All the calcium hydroxide formed was soluble.

$Ca(s) + 2H_2O(l) \longrightarrow Ca(OH)_2(aq) + H_2(g)$

The student transferred the contents of the beaker to a 250.0 cm^3 volumetric flask and water was added to make the solution up to 250.0 cm^3.

Calculate the concentration, in mol dm^{-3}, of hydroxide ions in the 250.0 cm^3 solution. 0.0105 mol dm^{-3}

(OCR Advanced Subsidary GCE, Chem. A, F321, Jan. 2011, 4(a), (b)(iii))

15 Outline the steps in preparing solid lead(II) sulphate from solid lead(II) nitrate. You have to state the additional chemical reagents that are required, but you need NOT mention the apparatus involved.

(Communication mark will be awarded.)

(HKDSE, Paper 1B, 2012, 6)

16 A green crystalline solid, D, dissolves in water to give a pale green solution.

a) When dilute sodium hydroxide solution is added to the solution of D, a green precipitate, E, is observed.

When aqueous ammonia is added to the solution of D, the same green precipitate, E, forms which does not dissolve in excess aqueous ammonia.

Identify, by name or formula, substance E.
Iron(II) hydroxide / $Fe(OH)_2$

b) When a solution of D is warmed with dilute sodium hydroxide solution, a gas, F, which turns moist red litmus paper blue, is given off.

Identify, by name or formula, the gas F.
Ammonia / NH_3

c) When a solution of barium chloride is added to a solution of D, a white precipitate, G, forms which is insoluble in dilute hydrochloric acid.

Identify, by name or formula, the precipitate G.
Barium sulphate / $BaSO_4$

d) Deduce the formulae of the three ions present in the salt, D. Fe^{2+}, NH_4^+, SO_4^{2-}

(Edexcel Advanced GCE, Unit 6B, Jan. 2012, 2)

17 An experiment was performed to determine the concentration of an ammonia solution. Firstly, 25.0 cm^3 of 2.0 M hydrochloric acid were diluted with distilled water to 250.0 cm^3. After that, 25.0 cm^3 of the diluted hydrochloric acid were titrated with the ammonia solution using methyl orange as the indicator. 22.90 cm^3 of the ammonia solution were required to reach the end point.

a) Name one piece of the glass apparatus that must be used in the dilution process.

b) Calculate the concentration of the diluted hydrochloric acid.

c) Draw a labelled diagram to show the set-up used in the titration.

d) State the expected colour change at the end point.

e) Write a chemical equation for the reaction involved.

f) Calculate the concentration of the ammonia solution.

(HKCEE, Paper 1, 2011, 9)

18 This question is about calcium hydroxide, $Ca(OH)_2$.
Refer to page T52.
The solubility of calcium hydroxide in water can be found by titrating a saturated solution of calcium hydroxide with hydrochloric acid of known concentration.

a) Describe how you would make a saturated solution of calcium hydroxide suitable for use in this titration. Do NOT describe the subsequent titration procedure.

b) 10.0 cm^3 portions of the saturated solution are placed in a conical flask and titrated with 0.0500 mol dm^{-3} hydrochloric acid added from a burette.

i) Name the apparatus used to measure the 10.0 cm^3 portions.

ii) Suggest a suitable indicator for this titration and state the colour change you would expect to see at the end point.

c) The following results were obtained:

Titration numbers	1	2	3
Final burette reading (cm^3)	19.20	28.05	37.10
Initial burette reading (cm^3)	10.00	19.20	28.15
Titre (cm^3)	9.20	8.85	8.95

i) Fill in the third titre value in the table.

ii) Suggest why the first titre should be disregarded.

iii) Calculate the mean titre.
8.90 cm^3

iv) Calculate the concentration of calcium hydroxide in mol dm^{-3}.
0.0223 mol dm^{-3}

v) Calculate the solubility of calcium hydroxide in g dm^{-3}.
1.65 g dm^{-3}
(Relative atomic masses: H = 1.0, O = 16.0, Ca = 40.1)

(Edexcel Advanced Subsidiary GCE, Unit 3B, Jan. 2010, 3(a)–(b), (c)(i)–(iii), (vi)–(vii))

19 The structure of a dibasic acid with chemical formula $H_2C_2O_4$ is shown below:



a) A student expected a 0.0500 mol dm^{-3} standard $H_2C_2O_4(aq)$ to have a pH of 1.0. However, the pH of the solution, when measured with a calibrated pH meter, was found to be greater than 1. Explain this observation with the aid of a chemical equation.

b) Solid sodium hydroxide is available in school laboratories. However, standard NaOH(aq) CANNOT be directly prepared by weighing NaOH(s) and then dissolving it in water. Explain why.

c) In a titration experiment, 25.00 cm^3 of a 0.0500 mol dm^{-3} standard $H_2C_2O_4(aq)$ and a few drops of phenolphthalein indicator were placed in a conical flask. NaOH(aq) of unknown concentration was then added from a burette into the flask. 17.20 cm^3 of the NaOH(aq) were required to reach the titration end point.

i) State the colour change at the titration end point.

ii) From the titration results, calculate the concentration of the NaOH(aq) in mol dm^{-3}.

d) The following were considered as INAPPROPRIATE practices when carrying out the experiment in (c). For each of them, explain why it would lead to inaccurate titration results:

i) rinsing the conical flask with the standard $H_2C_2O_4(aq)$ before transferring 25.00 cm^3 of the acid solution to it;

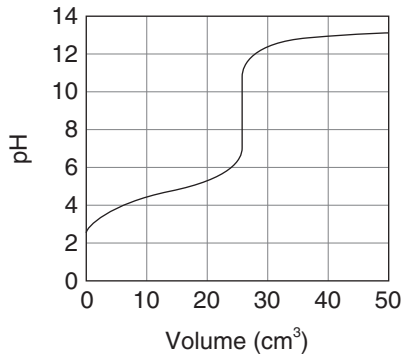
ii) carrying out the titration with the filter funnel remaining on top of the burette after using it to fill the burette with the NaOH(aq).

(HKDSE, Paper 1B, 2013, 4(b)–(e))

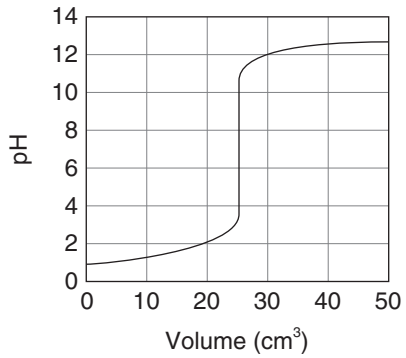
20 Titration curves labelled A, B, C and D for combinations of different aqueous solutions of acids and bases are shown below.

All solutions have a concentration of 0.1 mol dm^{-3}.

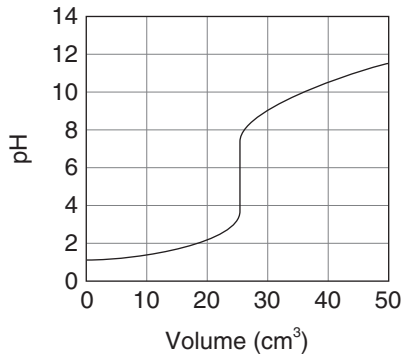
A



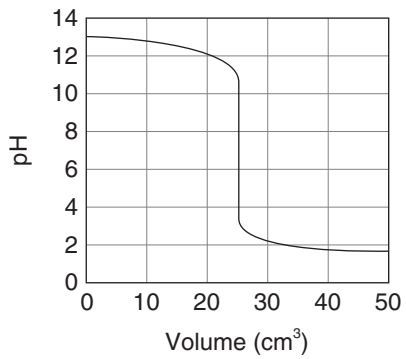
B



C



D



a) In this part of the question write the appropriate letter.

From the curves A, B, C and D, choose the curve produced by the addition of

i) aqueous ammonia to 25 cm^3 of hydrochloric acid;
C

ii) sodium hydroxide solution to 25 cm^3 of ethanoic acid; and
A

iii) nitric acid to 25 cm^3 of potassium hydroxide solution.
D

b) A table of acid-base indicators is shown below.

The pH ranges over which the indicators change colour and their colours in acid and alkali are also shown.

Indicator	pH range	Colour in acid	Colour in alkali
Trapaeolin	1.3–3.0	red	yellow
Bromocresol green	3.8–5.4	yellow	blue
Cresol purple	7.6–9.2	yellow	purple
Alizarin yellow	10.1–12.0	yellow	orange

i) Select from the table an indicator that could be used in the titration that produces curve B but NOT in the titration that produces curve A.
Bromocresol green

ii) Give the colour change at the end point of the titration that produces curve D when cresol purple is used as the indicator. Purple to yellow

(AQA Advanced GCE, Unit 4, Jun. 2011, 1)

21 Sodium tartrate and copper(II) nitrate are both salts.

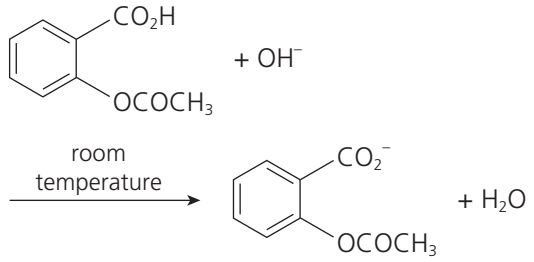
Sodium tartrate is a salt of tartaric acid. The formula of tartaric acid can be represented as H_xA. In this formula, x is the number of H^+ ions that can be replaced by metal ions to form salts.

A student carries out a titration to find the value of x in the formula of tartaric acid, H_xA. In the titration, 25.00 cm^3 of 0.0500 mol dm^{-3} tartaric acid, H_xA, exactly react with 12.50 cm^3 of 0.200 mol dm^{-3} sodium hydroxide solution, NaOH. A solution of sodium tartrate is produced.

Deduce the value for x in the formula of tartaric acid, H_xA. 2, refer to page T53.

(OCR Advanced Subsidiary GCE, Chem. A, F321, Jan. 2011, 2(a)(iii))

22 A commercial aspirin sample E was known to contain about 90% by mass of aspirin, while the rest was an inert binder. Based on the following reaction, a student designed an experiment and performed it at room temperature to determine the percentage by mass of aspirin in E.



The student added 2.25 g of E to 25.00 cm^3 of 3.05 mol dm^{-3} NaOH(aq), and then back titrated the excess NaOH(aq) with 2.50 mol dm^{-3} HCl(aq). The volume of HCl(aq) used was 23.10 cm^3.

a) Suggest an indicator for the titration.

b) From the student's experimental results, calculate the percentage by mass of aspirin in E.

(Relative molecular mass of aspirin = 180.0)

(HKASLE, Paper 1, 2012, 10(a)–(b))

23 A drug tablet contains aluminium hydroxide, $Al(OH)_3$, as the only active ingredient. A student performed the following experiment to determine the amount of aluminium hydroxide contained in the drug tablet.

Step	Experimental process	Remarks
I	A drug tablet was dissolved in 50.0 cm^3 of 1.0 M hydrochloric acid to form a solution.	As aluminium hydroxide is insoluble in water, the drug tablet was dissolved in hydrochloric acid instead. The amount of hydrochloric acid used was more than that needed to react with aluminium hydroxide in the drug tablet.
II	The solution was then diluted to 250.0 cm^3 with distilled water.	The solution, containing excess hydrochloric acid, was diluted for the titration in *Step III*.
III	25.0 cm^3 of the diluted solution was titrated with 0.20 M sodium hydroxide solution using a suitable indicator. 20.80 cm^3 of sodium hydroxide solution were needed to reach the end point.	The amount of excess hydrochloric acid in the diluted solution could be calculated from the data obtained in the titration.

a) Write a chemical equation for the reaction involved in *Step I*.

b) Describe how the dilution process in *Step II* should be performed by using suitable apparatus.

c) Suggest a suitable indicator for the titration in *Step III*, and state the expected colour change at the end point.

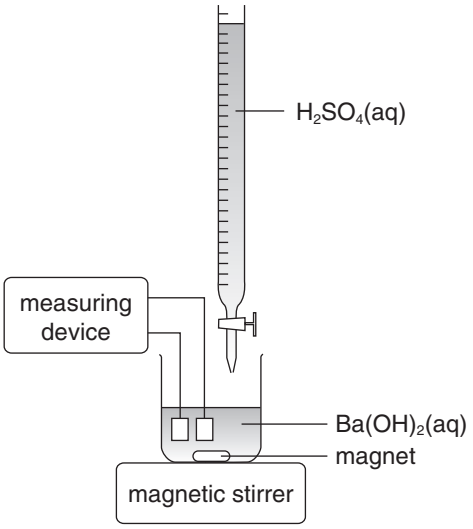
d) i) Calculate the number of moles of excess hydrochloric acid in the 25.0 cm^3 of the diluted solution from the data obtained in the titration.

ii) Hence, calculate the number of moles of aluminium hydroxide in the drug tablet.

(HKCEE, Paper 1, 2009, 11)

24 The diagram below shows the set-up of a titrimetric experiment involving the following reaction:

$Ba(OH)_2(aq) + H_2SO_4(aq) \longrightarrow BaSO_4(s) + 2H_2O(l)$



a) What physical parameter of the reaction mixture is measured by this set-up?

b) $H_2SO_4(aq)$ is added gradually to $Ba(OH)_2(aq)$ until in excess.

Sketch a graph to show the variation of the measured physical parameter with the volume of $H_2SO_4(aq)$ added. Explain your answer.

(HKASLE, Paper 1, 2010, 9)